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MINISTRY OF HIGHER EDUCATION

# INTERNSHIP REPORT

## DESIGN AND IMPLEMENTATION OF BOLO

### A WEB PLATFORM FOR RECRUITMENT AND JOB APPLICATION TRACKING

with customizable hiring stages and secure CV management

**Internship carried out at AFRILAND FIRST BANK, Douala, Bonanjo**

**From July 1 to August 31, 2026**

**Field: BACHELOR OF ENGINEERING (COMPUTER SCIENCE ENGINEERING).**


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**ACADEMIC YEAR 2026-2027**

# CERTIFICATION

This is to certify that the internship report entitled **"Design and Implementation of Bolo: a Web Platform for Recruitment and Job Application Tracking"** is the original work carried out by **SOLI MOKI Limnyuy Ngalla** at Afriland First Bank, Douala, from July 1 to August 31, 2026, in partial fulfilment of the requirements for the award of the Bachelor of Engineering (B.Eng.) degree in Computer Science Engineering at the School of Engineering and Applied Sciences (SEAS) of the Institut Universitaire de la Côte (IUC).

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| DATE…… | |

# DEDICATION

*To my family, for their love, their patience and their constant support.*

# ACKNOWLEDGEMENTS

Every achievement is the result of guidance, collaboration and perseverance. I am deeply grateful to all those whose support made this internship and this report possible.

First of all, I thank Almighty God for the life, the health and the strength granted to me throughout this internship. I also express my sincere gratitude to:

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- **The Director of the School of Engineering and Applied Sciences (SEAS) and the Head of the Computer Science Department**, for the quality of our training and their constant encouragement;
- **TANDAH MARCELLE**, my academic supervisor, for the availability, the advice and the careful review of this work;
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- **The staff of the Information Systems and Human Resources departments of Afriland First Bank**, Douala – Bonanjo, who explained their work, answered my questions on recruitment and tested the platform;
- **All the lecturers of the IUC**, for the knowledge and the method they passed on to us;
- **My family and my friends**, for their moral and material support at every stage of this work.

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# LIST OF ACRONYMS

| Acronym | Meaning |
|---|---|
| **ADCP** | Autorité de protection des données à caractère personnel (Cameroon) |
| **AES-GCM** | Advanced Encryption Standard – Galois/Counter Mode |
| **API** | Application Programming Interface |
| **ATS** | Applicant Tracking System |
| **BPMN** | Business Process Model and Notation |
| **CNPS** | Caisse Nationale de Prévoyance Sociale |
| **CV** | Curriculum Vitae |
| **FCFA** | Franc de la Coopération Financière en Afrique centrale |
| **FR / NFR** | Functional / Non-Functional Requirement |
| **GIN** | Generalized Inverted Index (PostgreSQL) |
| **HMAC** | Hash-based Message Authentication Code |
| **HR** | Human Resources |
| **HTTP(S)** | HyperText Transfer Protocol (Secure) |
| **IUC** | Institut Universitaire de la Côte |
| **JSON** | JavaScript Object Notation |
| **JWT** | JSON Web Token |
| **MoSCoW** | Must, Should, Could, Won't (prioritisation method) |
| **OWASP** | Open Worldwide Application Security Project |
| **RBAC** | Role-Based Access Control |
| **REST** | Representational State Transfer |
| **SEAS** | School of Engineering and Applied Sciences |
| **SLA** | Service Level Agreement (here: maximum time in a stage) |
| **SME** | Small and Medium-sized Enterprise |
| **SMS / SMTP** | Short Message Service / Simple Mail Transfer Protocol |
| **SPA** | Single-Page Application |
| **SUS** | System Usability Scale |
| **SYSCOHADA** | Système comptable de l'OHADA |
| **UML** | Unified Modeling Language |

# ABSTRACT

This report presents the work carried out during my internship at **Afriland First Bank**, Douala – Bonanjo, from July 1 to August 31, 2026, as part of the Bachelor of Engineering programme in Computer Science Engineering at the Institut Universitaire de la Côte (IUC).

Small and medium-sized enterprises, which form a large share of the bank's customers, receive CVs through many channels at once – WhatsApp, e-mail, paper copies dropped at the office, social networks – and lose track of applications. CVs are lost or duplicated, nobody has a shared view of where each candidate stands, candidates rarely receive an answer, and personal data are kept without protection on phones and shared computers.

To answer this problem I designed and built **Bolo** ("bolo" means "work" in Cameroonian street French), a web platform for recruitment and job application tracking. Bolo gathers every channel into one pipeline, models the recruitment process as a **configurable state machine** whose stages, delays and messages are defined per job offer, and manages CVs securely: type checked from the file content, antivirus scan, AES-256-GCM encryption at rest, five-minute signed links, an audit log of every view, and the right to erasure required by Law No. 2024/017 on personal data protection. Candidates apply from a light, bilingual (French/English) career page and always receive an answer in their language.

The project followed an iterative and incremental process: a requirements specification of 34 functional and 16 measurable non-functional requirements, a BPMN and UML design, and three tested increments on a React, Node.js/Express and PostgreSQL stack. The platform is validated by **43 automated integration tests (43 passed)**; on 10,000 candidates, a search answers in 39 ms (median) and moving a candidate takes 6 ms on the test server.

**Keywords:** recruitment, applicant tracking system, business process modelling, state machine, secure file management, personal data protection, React, Node.js, PostgreSQL, Cameroon.

# RÉSUMÉ

Ce rapport présente le travail réalisé pendant mon stage chez **Afriland First Bank**, Douala – Bonanjo, du 1er juillet au 31 août 2026, dans le cadre du Bachelor of Engineering en Computer Science Engineering de l'Institut Universitaire de la Côte (IUC).

Les petites et moyennes entreprises, qui représentent une grande partie de la clientèle de la banque, reçoivent des CV par plusieurs canaux à la fois – WhatsApp, e-mail, CV papier déposés au bureau, réseaux sociaux – et perdent le suivi des candidatures. Des CV sont perdus ou en double, personne n'a une vue commune de l'avancement de chaque candidat, les candidats reçoivent rarement une réponse, et les données personnelles sont conservées sans protection.

Pour répondre à ce problème, j'ai conçu et développé **Bolo** (« bolo » signifie « travail » en camfranglais), une plateforme web de recrutement et de suivi des candidatures. Bolo réunit tous les canaux dans un même pipeline, modélise le processus de recrutement comme un **automate à états configurable** dont les étapes, les délais et les messages sont définis pour chaque offre, et gère les CV de façon sécurisée : type vérifié d'après le contenu du fichier, analyse antivirus, chiffrement AES-256-GCM, liens signés valables cinq minutes, journal de chaque consultation et droit à l'effacement prévu par la loi n° 2024/017 sur la protection des données personnelles. Les candidats postulent depuis une page carrière légère et bilingue et reçoivent toujours une réponse dans leur langue.

Le projet a suivi une démarche itérative et incrémentale : un cahier des charges de 34 exigences fonctionnelles et 16 exigences non fonctionnelles mesurables, une conception BPMN et UML, et trois incréments testés sur une pile React, Node.js/Express et PostgreSQL. La plateforme est validée par **43 tests d'intégration automatisés (43 réussis)** ; sur 10 000 candidats, une recherche répond en 39 ms (médiane) et le déplacement d'un candidat prend 6 ms sur le serveur de test.

**Mots-clés :** recrutement, suivi des candidatures, modélisation de processus métier, automate à états, gestion sécurisée des fichiers, protection des données personnelles, React, Node.js, PostgreSQL, Cameroun.

# GENERAL INTRODUCTION

## *General context of the study*

In Cameroon, small and medium-sized enterprises (SMEs) make up the large majority of companies and of formal jobs. They hire regularly – an accountant, sales agents, drivers, a storekeeper – but rarely have a Human Resources department, and never a recruitment software. A job offer is posted on Facebook, shared in WhatsApp groups and pinned at the entrance; applications then arrive through every channel at once: the personal WhatsApp of the HR officer, several mailboxes, paper CVs left at the reception, messages under the Facebook post. The same person may apply three times; another CV is never seen again.

As part of the Bachelor of Engineering programme in Computer Science Engineering at the School of Engineering and Applied Sciences (SEAS) of the Institut Universitaire de la Côte (IUC), I carried out my internship in the Information Systems department of **Afriland First Bank**, Douala – Bonanjo, from July 1 to August 31, 2026. SMEs are among the main customers of the bank, and the bank itself recruits staff and interns every year. Discussions with the HR team and with SME account managers showed that the same difficulty appears on both sides: **companies receive CVs through several channels and lose track of applications**. This led to the project **Bolo**, a web platform for recruitment and job application tracking.

## *Definition of the main concepts*

- **Applicant Tracking System (ATS):** software that centralises job offers and applications and follows each candidate until a decision.
- **Recruitment pipeline:** the ordered list of stages (for example New, Screening, Interview, Test, Offer) through which every application moves until it is hired or rejected.
- **Business process modelling:** the representation of the activities, actors, rules and events of a business process (here, recruitment) with a notation such as BPMN, in order to analyse and automate it.
- **State machine:** a model in which an object (an application) is always in exactly one state (a stage) and changes state only through allowed transitions triggered by events.
- **Secure CV management:** the technical and organisational measures that keep the CVs confidential, intact and available to authorised people only, and processed lawfully.
- **Personal data:** any information relating to an identified or identifiable person – a CV contains many of them (identity, phone, career, sometimes a photo).

## *Problematic*

The question is not whether a recruitment software can be written – many exist – but whether one can be designed for the reality of a Cameroonian SME: CVs arriving through WhatsApp and on paper, candidates who give a phone number before an e-mail address, two official languages, prepaid mobile data, a recruitment practice that differs from one company and one position to another, and a new law

on personal data protection. The central question is therefore: **how can we design and implement, with one developer, in nine weeks and at a very low cost, a web platform that gathers applications from every channel into one traceable process, adapts to the recruitment stages of each company, and protects the CVs it stores?**

### *Aim of the study*

The general objective is to design and implement Bolo. The specific objectives are to: (1) offer a single entry point for all applications – an online career page and a quick capture of the CVs received elsewhere; (2) model the recruitment process so that each company, and each offer, can define its own stages, delays and messages; (3) let HR staff find the right candidates quickly through search and filters, including in the text of the CVs; (4) keep candidates and the team informed with automatic, bilingual messages and notifications, so that no candidate is left without an answer; (5) store and show CVs securely, in line with Law No. 2024/017; and (6) apply a rigorous method with measurable requirements that become tests. For Afriland First Bank, the project is a concrete tool for its SME customers and for its own recruitment of interns.

### *Organisation of the work*

Besides this introduction and the general conclusion, the report has two parts. **Part One (conceptual framework)** presents the host company and the internship activities (Chapter 1) and the literature review (Chapter 2). **Part Two (empirical study)** analyses the existing system and specifies the requirements (Chapter 3), then presents the solution, its implementation, tests, installation and cost (Chapter 4).

# PART ONE:
# CONCEPTUAL FRAMEWORK

# CHAPTER 1
# PRESENTATION OF THE COMPANY AND ACTIVITIES CARRIED OUT DURING THE INTERNSHIP

## Introduction

This chapter presents Afriland First Bank, the host company of the internship, then the activities carried out week by week, and finally the choice of the topic.

## 1.1 Presentation of the company

### *1.1.1 Identification sheet*

*Table 1: Identification sheet of Afriland First Bank*

| Item | Information |
|---|---|
| **Logo** | (see cover page) |
| **Company name** | Afriland First Bank |
| **Original name** | Caisse Commune d'Épargne et d'Investissement (CCEI) |
| **Legal form** | Société Anonyme (S.A.) – commercial bank |
| **Creation** | 1987 (first counter operational on July 1, 1988) |
| **Head office** | Yaoundé, Hippodrome district |
| **Network** | About forty branches in the ten regions of Cameroon |
| **Internship site** | Douala – Bonanjo, Information Systems department |
| **Size** | 500 to 1,000 employees (public directory) |
| **Website** | www.afrilandfirstbank.com |

### *1.1.2 History and mission*

The bank was created in 1987 under the name **Caisse Commune d'Épargne et d'Investissement (CCEI)** and opened its first counter in Yaoundé on July 1, 1988, at a time when many banks operating in the country were failing. From the start, its customers were mainly **SMEs and individuals**, and supporting local entrepreneurs remains at the heart of its identity. The institution later took the name Afriland First Bank and developed a network of about forty branches across the ten regions of the country. Its mission is to offer banking services adapted to Cameroonian companies and households and to finance the local economy.

### *1.1.3 Business lines*

*Table 2: Main business lines of the bank*

| Business line | Content |
|---|---|
| **Retail banking** | Current and savings accounts, cards, loans to individuals |
| **SME and corporate banking** | Business accounts, credit, cash management, advice to entrepreneurs |

| Business line | Content |
|---|---|
| **Payments and transfers** | National and international transfers, trade finance |
| **Digital banking** | Mobile and internet banking, electronic channels |
| **Information systems** | Core banking, networks, security and internal applications (internship unit) |

### *1.1.4 Organisation*

The General Management, based at the head office in Yaoundé, coordinates functional departments and the branch network (Figure 1). The Information Systems department ensures the availability and security of the information system and develops internal applications for the other departments. I was attached to its **applications and development unit** in Douala – Bonanjo, which allowed me to work both with developers and with the business users of the Human Resources department.

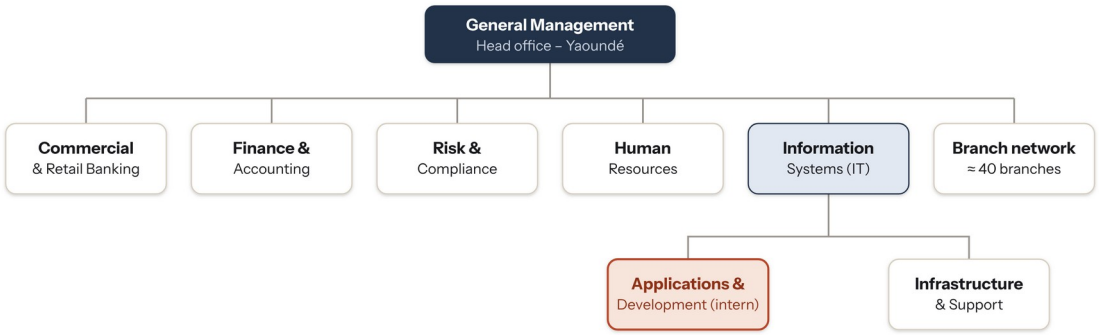



***Figure 1:*** *Simplified organisation chart of Afriland First Bank (source: author)*

## 1.2 Activities carried out during the internship

The internship started on Wednesday, July 1, 2026. I was received by my professional supervisor, who presented the bank, its internal rules (confidentiality, use of IT resources, working hours) and the team. After a first week of discovery and help-desk support, the subject was agreed upon: a tool that gathers and follows job applications, useful to the bank's SME customers and to its own recruitment of interns. The nine weeks then followed the phases of the software development life cycle (Figure 2 and Table 3).

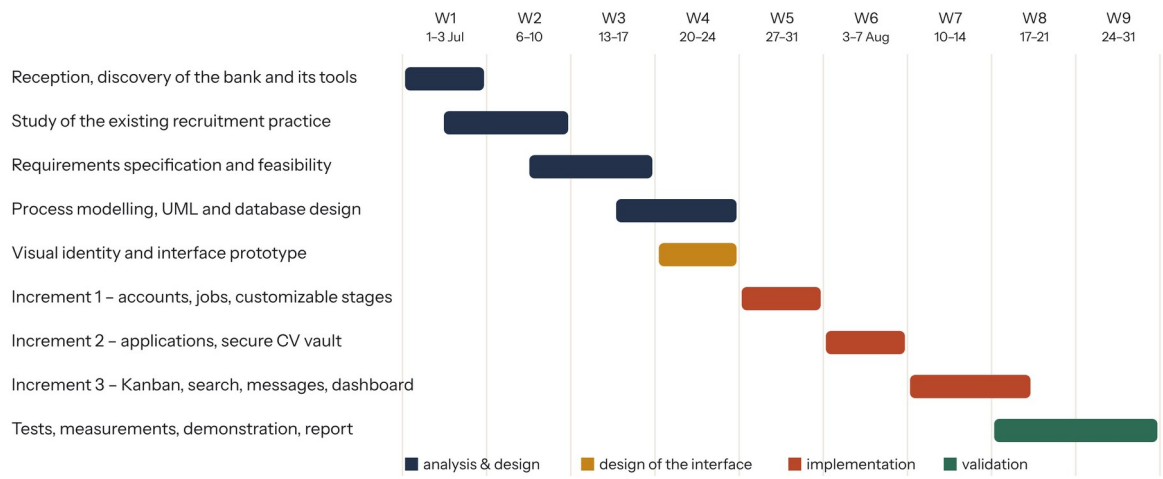



***Figure 2:*** *Planning of the internship (July 1 – August 31, 2026) (source: author)*

***Table 3:*** *Weekly summary of the activities carried out*

| Week | Activities carried out | Lessons and problems solved |
|---|---|---|
| 1 (Jul 1–3) | Reception, discovery of the bank, its tools and security rules; help-desk support | How a bank organises its information system |
| 2 (Jul 6–10) | Interviews with the HR team and two SME account managers; collection of Excel sheets and e-mails | Understand the business need before coding |
| 3 (Jul 13–17) | Requirements specification (34 FR, 16 NFR); feasibility study; choice of the process model | Writing measurable requirements that become tests |
| 4 (Jul 20–24) | BPMN process model, UML diagrams, database design; visual identity and interface prototype | Stages must be data, not code |
| 5 (Jul 27–31) | Increment 1: project setup, authentication, roles, job offers, customizable stages | JWT in httpOnly cookie, bcrypt, Zod validation |
| 6 (Aug 3–7) | Increment 2: online application, CV vault (type check, scan, encryption), manual capture | File upload security (OWASP); no orphan files |
| 7 (Aug 10–14) | Increment 3: Kanban pipeline, business rules, bilingual messages, search in CV text | Transactions; PostgreSQL full-text search |
| 8 (Aug 17–21) | Increment 3 (end): dashboard, audit log, erasure and retention; integration tests | A test per business rule; one clock for time rules |
| 9 (Aug 24–31) | Measurements on 10,000 candidates, demonstration to HR, documentation, report | Reporting honestly what is and is not verified |

### *1.2.1 Detail of the main activities*

**Weeks 1 and 2 – discovery and framing.** Besides help-desk tasks (workstation configuration, network incidents), I interviewed the HR officer in charge of the recruitment of interns and two account managers in charge of SME customers. I collected anonymised examples of what they use today: an Excel sheet per offer, CVs forwarded by e-mail, WhatsApp conversations. Three observations shaped the project: the **phone number** is the identifier every candidate has (many give no e-mail), **paper and WhatsApp CVs** must enter the same process as online applications, and each company – even each position – has **different stages**.

**Weeks 3 and 4 – analysis and design.** The requirements were written with MoSCoW priorities and measurable targets (Chapter 3); the feasibility study showed a direct cost limited to hosting. The recruitment process was modelled in BPMN and as a configurable state machine, then the architecture, the data model and the REST API were designed. A complete visual identity (name, logo, colours, fonts) was created so that the product would not look like a generic template.

**Weeks 5 to 9 – implementation and tests.** Each increment delivered a working version tested by automated integration tests against a real PostgreSQL database. The last two weeks were devoted to measurements, a demonstration to the HR team, documentation and this report.

This internship strengthened my technical skills (React, Node.js, Express, PostgreSQL, application security, Git) and my methodological skills (requirements engineering, business process modelling, UML, test planning, technical writing).

## 1.3 Choice and justification of the topic

The topic **“Design and implementation of Bolo, a web platform for recruitment and job application tracking”** was chosen because it answers a real and frequent problem of the bank’s SME customers and of its own HR team; it is aligned with the activities of the Information Systems department (internal applications, security of data); it is a complete engineering exercise – process modelling, web development, database, security, tests – and it is feasible by one developer in nine weeks with free and open-source tools.

## Conclusion

This chapter presented Afriland First Bank, a Cameroonian bank close to SMEs, the activities of the nine weeks of internship and the reasons for the choice of the topic. The next chapter reviews the concepts and technologies needed for the study.

# CHAPTER 2
# OVERVIEW OF THE TOPIC STUDIED (LITERATURE REVIEW)

## Introduction

This chapter presents the theory needed to understand the study: the recruitment process, business process modelling, the protection of personal data and the security of uploaded files, the architecture of web applications, the existing solutions and the software process models.

## 2.1 The recruitment process

Whatever the size of the company, a recruitment follows the same main phases: definition of the need, sourcing (publication of the offer), collection of applications, selection, decision and offer, integration, and closure – answering every candidate who was not selected. The **selection** phase varies most: a bank may require several interviews and background checks, a small shop may hire after one interview, a transport company adds a driving test. A useful tool must therefore let each company define its own stages, and even adapt them for one offer.

Two weaknesses are frequent in SMEs: the **collection** phase is scattered over many channels, and the **closure** phase is often forgotten – candidates who were not selected never hear back, which harms the image of the employer.

## 2.2 Business process modelling

A business process is a set of activities, performed by actors, that turns an input into a result of value. Business Process Management (Dumas et al., 2018) studies how to identify, model, analyse, automate and improve such processes. Several notations exist (Table 4).

***Table 4:*** *Notations for modelling the recruitment process*

| Notation | Strengths | Use in this project |
|---|---|---|
| BPMN 2.0 | Standard of the OMG; lanes per actor; events and gateways; readable by business users | Global process with the candidate, the system, HR and recruiters |
| UML activity diagram | Close to BPMN; integrated with the other UML diagrams | Not used (BPMN preferred for the business view) |
| UML state machine | Describes the life cycle of one object: states, transitions, guards | Life cycle of an application; basis of the implementation |

The recruitment process lends itself well to a **state machine**: each application is an object whose state is its current stage; the events (move, hire, reject) are transitions performed by authorised actors; guards express the rules (a rejection needs a reason); actions express the effects (a message to the candidate, a line in the history). If the states are stored as **data** rather than written in the code, every

company can change them without a new version of the software – this is the key design choice of Bolo.

## 2.3 Protection of personal data and security of CVs

A CV contains personal data. In Cameroon, **Law No. 2024/017 of 23 December 2024** on the protection of personal data applies to public and private processing; it gives data subjects the rights of access, rectification, deletion, objection and portability, restricts transfers of data abroad, and creates a supervisory authority, the ADCP. The European GDPR follows the same principles. For a recruitment tool they mean: inform the candidate and obtain consent; collect only what is necessary; keep the data for a limited time; protect it against loss and unauthorised access; and allow its deletion.

On the technical side, the OWASP File Upload Cheat Sheet recommends: an allow-list of file types checked from the **content** (magic bytes) and not from the extension; a size limit; a random file name; storage outside the web root; an antivirus scan; and access through authorised, short-lived links. Encrypting files at rest with an authenticated algorithm such as **AES-256-GCM** (NIST SP 800-38D) protects them even if the disk or a backup is stolen.

## 2.4 Architecture of web applications

A single-page application (SPA) loads once in the browser and then exchanges only data with the server through a **REST** API (Fielding, 2000). **React** builds the interface from reusable components; **Node.js** runs JavaScript on the server with an event loop suited to many simultaneous requests; **Express** is its most used web framework. For the data, the choice is between a relational and a document database (Table 5).

*Table 5: Relational and document-oriented databases for recruitment data*

| Criterion | Relational (PostgreSQL) | Document (MongoDB) |
|---|---|---|
| Data model | Tables linked by foreign keys | JSON-like documents |
| Integrity | Constraints (unique, foreign keys, check) | Mostly checked by the application |
| Transactions | Full ACID transactions | Multi-document transactions possible, less natural |
| Search | Native full-text search (tsvector, GIN index) | Text index |
| Fit for this project | Applications, stages, history and rules are strongly related | Less suited to the many relations |

PostgreSQL was chosen: recruitment data are strongly related (a job has stages, an application points to a stage, a move creates a history line), and a stage change must update several tables **in one transaction**. Its full-text search also allows searching inside the text of the CVs without an extra search engine.

## 2.5 Existing solutions

*Table 6: Existing solutions and the needs of the study*

| Need | Excel + WhatsApp | Google Forms + Sheets | International ATS (SaaS) | Open-source ATS | Bolo |
|---|---|---|---|---|---|
| One entry for all channels | No | Online only | Online + e-mail | Partly | Online + manual capture |
| Stages per offer | No | No | Yes | Limited | Yes, with delay and message |
| Paper / photo CV | Manual | No | Rarely | No | Yes (photo resized) |
| Phone as identifier | Yes | No | No (e-mail) | No | Yes |
| FR / EN answers | Manual | No | Yes, paid plans | Partly | Automatic, per candidate |
| CV security | None | Third-party storage | Good | Variable | Encryption, signed links, log |
| Cost for an SME | Free | Free | Monthly fee per user, card payment | Free, hard to host | Hosting only |

International ATS cover most needs but cost a monthly fee per user in foreign currency, are designed around e-mail and do not take paper or WhatsApp CVs into account; free tools do not model a process nor protect the data. None combines these needs for a Cameroonian SME.

## 2.6 Software process models

*Table 7: Comparison of software process models*

| Model | Advantages | Limits for this project |
|---|---|---|
| Waterfall | Simple, well documented | Working product only at the end |
| V-model | Tests designed early | Sequential, little room for change |
| Iterative and incremental | A working version after each increment; risks isolated | Needs a stable architecture |
| Agile (Scrum) | Very flexible | Rituals heavy for one developer |

The **iterative and incremental** model was chosen: the requirements were mostly known after the interviews, the deadline was strict, and the riskiest part (secure file handling) could be isolated in its own increment. From the V-model, the measurable non-functional requirements were reused directly as acceptance tests.

## Conclusion

This chapter presented the recruitment process, BPMN and state machines, the legal and technical rules for protecting CVs, the web architecture and PostgreSQL, the existing solutions and the iterative model, and showed that no existing solution fits the needs of a Cameroonian SME.

# PART TWO:
# EMPIRICAL STUDY

# CHAPTER 3
# DIAGNOSTIC OF THE EXISTING SYSTEM AND REQUIREMENTS SPECIFICATION

## Introduction

This chapter presents the existing system – the way applications are received and followed today – its functions, advantages and limits, and derives the requirements specification (cahier des charges) and the feasibility of the new system.

## 3.1 Presentation and functions of the existing system

The existing system is not a software but a set of practices around general-purpose tools (Figure 3). Applications arrive through five channels; the HR officer downloads, prints or scans them, records the candidates in an Excel sheet per offer, keeps the files in a shared folder and forwards some CVs to the managers by e-mail; interviews are arranged by phone.

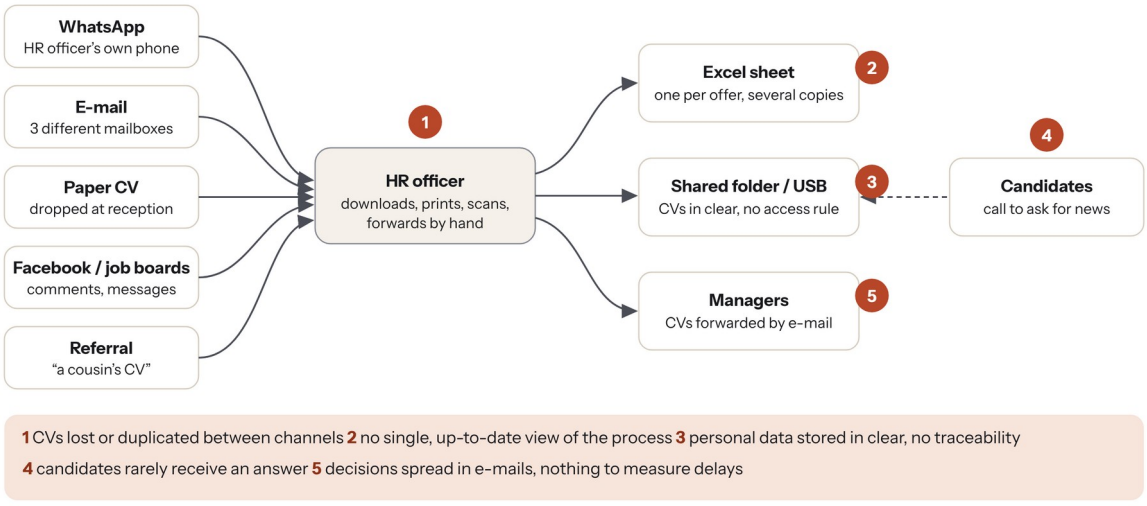



*Figure 3: Topology of the existing situation and its five main problems (source: author)*

*Table 8: Functions of the existing system*

| Function | How it is done today |
|---|---|
| **Publish an offer** | Word document shared on Facebook, WhatsApp groups and at the entrance |
| **Receive applications** | WhatsApp, three mailboxes, paper, social networks, referrals |
| **Record candidates** | One Excel sheet per offer, copied and sent around |
| **Select** | CVs forwarded by e-mail to managers; comments in replies |
| **Inform candidates** | Phone call to the selected candidate; others rarely informed |

## 3.2 Advantages of the existing system

**Technically**, the tools are already known and available: everybody uses WhatsApp and Excel, nothing must be installed and the process adapts to any situation. **Financially**, it has no direct cost.

## 3.3 Limits of the existing system

*Table 9: Limits of the existing system and their consequences*

| Limit | Observed situation | Consequence |
|---|---|---|
| **Scattered channels** | Five channels, three mailboxes, a personal phone | CVs lost or duplicated; good profiles missed |
| **No shared view** | Several copies of the Excel sheet | Nobody knows the real state of a candidate |
| **No traceability** | Decisions in e-mails and phone calls | Delays cannot be measured; disputes cannot be checked |
| **Candidates without answer** | Only the selected candidate is called | Bad image of the employer; repeated calls to HR |
| **Unprotected personal data** | CVs on phones, USB keys, shared folders | Risk of leaks; not compliant with Law No. 2024/017 |
| **Hidden cost** | Hours of copying, searching and calling | Slow recruitments, sometimes a poor hire |

The new system must overcome these limits while keeping the advantages of the existing practice: candidates must still be able to use WhatsApp or a paper CV, and the cost for the company must stay very low.

## 3.4 Requirements specification (cahier des charges)

### *3.4.1 Actors and use cases*

The system has five human actors – Visitor, Candidate (a Visitor who applies), Recruiter (a line manager assigned to some offers), HR manager (a Recruiter with all rights on the company's offers) and Administrator – and two external systems: the e-mail/SMS gateway and the antivirus. Figure 4 shows the use cases; the highlighted ones are the signature features of Bolo.

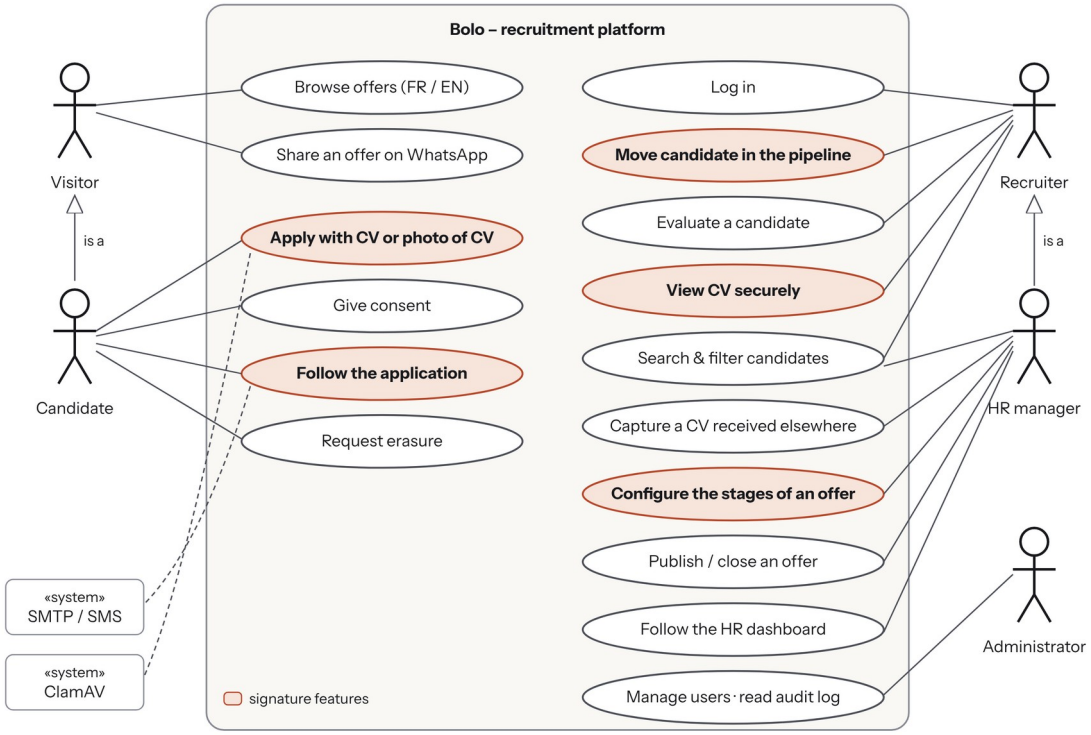



*Figure 4: Use case diagram of Bolo (source: author)*

### 3.4.2 Functional requirements

The 34 functional requirements are grouped in seven modules and prioritised with the MoSCoW method (Table 10).

*Table 10: Functional requirements by module (M = Must, S = Should, C = Could)*

| Module | Main requirements | Priorities |
|---|---|---|
| **Accounts & access (FR-01–04)** | Log in / out; roles admin, HR, recruiter; lock after failed attempts; user management | 3 M, 1 S |
| **Job offers (FR-05–09)** | Create and edit; publish on the career page; close with an answer to every candidate; share on WhatsApp; assign recruiters | 3 M, 2 S |
| **A. Configurable process (FR-10–14)** | Stages per offer (add, rename, reorder, delete); Kanban moves; history of every move; maximum delay per stage; rejection reason and delayed answer | 4 M, 1 S |
| **B. Applications & CVs (FR-15–21)** | Online application with CV or photo; capture of CVs received by WhatsApp, e-mail, paper or referral; no duplicate (phone); secure viewing; PDF, Word, JPG, PNG; consent; photo resized in browser | 6 M, 1 C |
| **Search (FR-22–24)** | Full-text search in profiles and CV text; filters (offer, stage, experience, source, language, rating); evaluations 1–5 | 2 M, 1 S |
| **C. Communication (FR-25–29)** | Bilingual templates; e-mail or SMS when no e-mail; in-app notifications; public tracking page; real sending through SMTP / SMS gateway | 1 M, 4 S |
| **D. Dashboard & compliance (FR-30–34)** | HR dashboard; audit log of CV views; right to erasure; automatic anonymisation after 24 months; export to Excel; interview scheduling | 3 M, 1 S, 2 C |

### *3.4.3 Non-functional requirements*

Each of the 16 non-functional requirements has a measurable target, so that it becomes an acceptance test (Table 11).

*Table 11: Non-functional requirements and measurable targets*

| ID | Category | Target |
|---|---|---|
| **NFR-01** | Performance | Search among 10,000 candidates in under 1 s |
| **NFR-02** | Performance | Moving a candidate answered in under 300 ms |
| **NFR-03** | Performance / data | Career page under 300 kB transferred; usable on 3G |
| **NFR-04** | Performance | Application with a 1 MB CV processed by the server in under 2 s |
| **NFR-05** | Security | Passwords hashed with bcrypt (cost 12), never returned |
| **NFR-06** | Security | Account locked 15 min after 5 failed logins |
| **NFR-07** | Security | Every private route requires a valid session and the right role |
| **NFR-08** | Security | Strict isolation between companies |
| **NFR-09** | Security | CV type checked from content, antivirus scan, AES-256-GCM at rest |
| **NFR-10** | Security | CV links signed, bound to one user, valid 5 minutes |
| **NFR-11** | Traceability | Every CV view written in the audit log |
| **NFR-12** | Privacy | Consent required; erasure on request; anonymisation after 24 months |
| **NFR-13** | Usability | A new HR user publishes an offer in under 10 min; SUS ≥ 70 |
| **NFR-14** | Compatibility | Recent Chrome, Firefox, Edge, Safari; from 360 px wide |
| **NFR-15** | Maintainability | Layered code; automated tests for every business rule |
| **NFR-16** | Availability | Installation with one command; daily database backup |

### *3.4.4 Constraints and deliverables*

- **Technical:** React for the client, Node.js/Express for the server, PostgreSQL for the database (stack set by the project).
- **Time:** nine weeks, one developer.
- **Budget:** free and open-source tools; low-cost hosting suited to an SME.
- **Legal:** compliance with Law No. 2024/017 on personal data protection.
- **Deliverables:** requirements specification, design, source code with tests, installation files, user guide and this report.

## 3.5 Feasibility study

**Economic feasibility.** All the software used is free and open source (React, Node.js, PostgreSQL, ClamAV, Docker). The only recurring cost is the hosting of a small server, a domain name and the sending of e-mails and SMS (section 4.9). **Technical feasibility.** The stack is mastered in our training; the new elements – full-text search, file encryption, antivirus – are documented and testable locally. **Legal feasibility.** Only the data needed for recruitment are collected; consent is recorded; data stay on a server chosen by the company; candidates can obtain the deletion of their data. **Operational**

**feasibility.** The HR team keeps its habits: a CV received on WhatsApp is simply captured in Bolo with its source.

*Table 12: Main technical risks and mitigations*

| Risk | Mitigation |
|---|---|
| **Malicious file uploaded as a CV** | Type checked from content, size limit, antivirus, file never executed nor served directly |
| **Loss of the encryption key** | Key kept outside the database and the code; offline copy documented in the installation guide |
| **Candidates without e-mail** | Phone number as identifier; SMS as fallback channel |
| **Stages changed while candidates are in them** | A stage that holds applications cannot be deleted (rule BR5) |
| **Poor connection of candidates** | Light public pages loaded separately; photos resized in the browser |

## Conclusion

The existing practice, although free and familiar, scatters applications, leaves no trace, keeps CVs unprotected and leaves candidates without answer. These limits were turned into 34 functional and 16 measurable non-functional requirements, and the new system is feasible at a very low cost.

# CHAPTER 4
# PROPOSED SOLUTION AND IMPLEMENTATION

## Introduction

This chapter presents Bolo: the technical notebook, the architecture, the modelling of the recruitment process, the detailed design, the security of CVs, the user interface, the implementation, the tests and results, the installation procedure and the cost estimate.

## 4.1 Technical notebook of the new system

*Table 13: Technologies chosen and justification*

| Layer | Technology | Justification |
|---|---|---|
| **Interface** | React 18, Vite 5, React Router 6 | Reusable components; fast build; public pages and back-office in separate bundles |
| **CV display** | pdf.js, mammoth | CVs shown inside the application with a watermark, without download button |
| **Server** | Node.js 20, Express 4, Helmet | Event loop suited to uploads and many requests; secure HTTP headers |
| **Validation** | Zod | Every request body and query checked against a schema |
| **Database** | PostgreSQL 16 (driver pg) | Transactions, constraints, arrays, full-text search with GIN indexes |
| **Security** | jsonwebtoken, bcryptjs, express-rate-limit, Node crypto | Session in an httpOnly cookie; hashed passwords; limits; AES-256-GCM and HMAC |
| **Files** | Multer (memory), ClamAV | Upload limited to 5 MB; antivirus scan before storage |
| **Tests** | node:test, Supertest | Integration tests through the HTTP API against a real PostgreSQL database |
| **Deployment** | Docker Compose, Nginx | Installation with one command; compression and cache of static files |

## 4.2 Architecture of the new system

Bolo uses a three-tier client-server architecture (Figure 5). The browser loads either the **HR back-office** or the **public career pages**; the two are separate bundles, so that a candidate on a 3G connection never downloads the back-office. The Express server checks the session and the role, applies the business rules in a service layer and accesses PostgreSQL, the encrypted CV storage and the antivirus. Messages to candidates are written in an **outbox** table in the same transaction as the decision, then sent by a background job – a message is therefore never lost, nor sent for a decision that was cancelled.

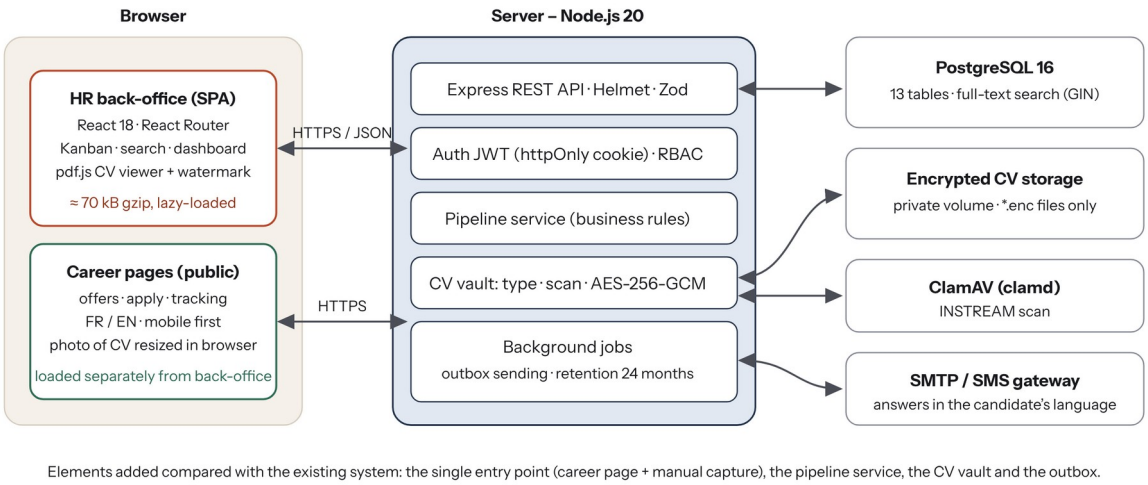



***Figure 5:*** *High-level architecture of Bolo (source: author)*

Figure 6 shows where each part runs in production, and Figure 7 the layered organisation of the server code and of the repository.

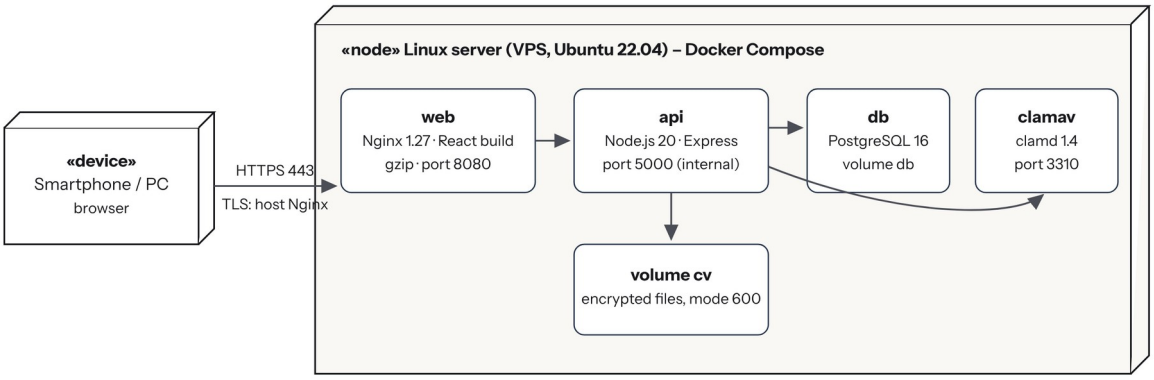



***Figure 6:*** *Deployment diagram (source: author)*

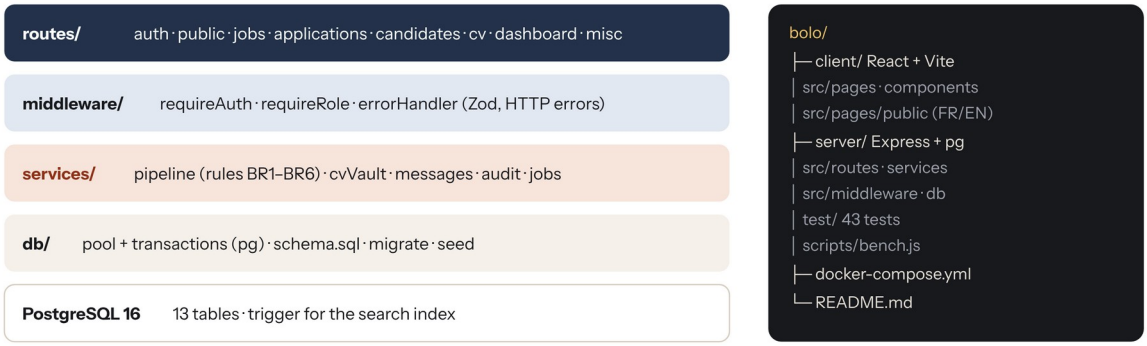



***Figure 7:*** *Layered structure of the server and organisation of the repository (source: author)*

## 4.3 Modelling of the recruitment business process

The recruitment process was first modelled in **BPMN 2.0** with one lane per actor (Figure 8). It shows the single entry point (the candidate applies, or HR captures a CV received elsewhere), the checks done by the system before the application exists, the work of HR and recruiters, and the end of the process – a message in the candidate's language, whatever the decision.

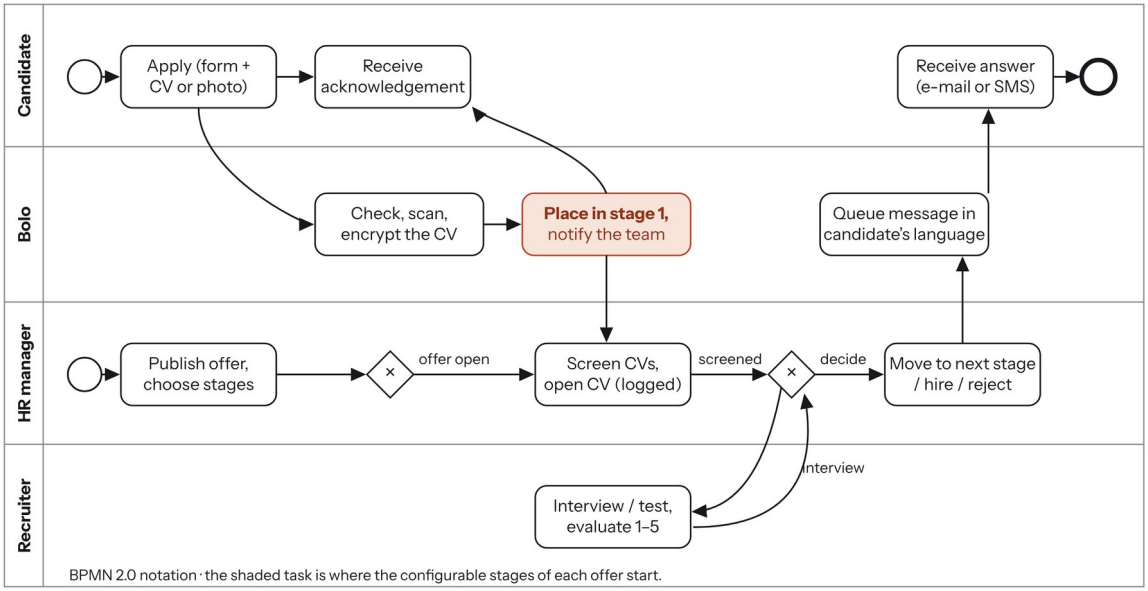



*Figure 8: BPMN model of the recruitment process with Bolo (source: author)*

The life cycle of an application is then a **configurable state machine** (Figure 9). The states are rows of the table `stages`: when an offer is created, the company template is copied, and HR can rename, add, remove or reorder the stages of that offer only, and set for each one a maximum delay (SLA) and the message sent to the candidate. The rules of Table 14 are enforced by the service layer and each of them is covered by an automated test.

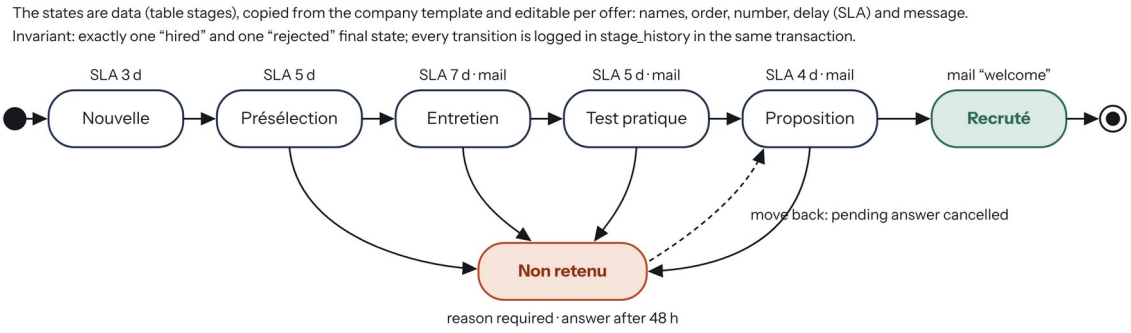



*Figure 9: State machine of an application (default stages of the demonstration company) (source: author)*

*Table 14: Business rules of the recruitment pipeline*

| Rule | Description | Test |
|---|---|---|
| **BR1** | An application can only move to a stage of its own offer | pipeline #4 |
| **BR2** | Admins and HR manage all offers; a recruiter only the offers assigned to them | pipeline #7 |

| Rule | Description | Test |
|---|---|---|
| **BR3** | A rejection needs a reason; the answer is sent 48 h later and can be cancelled | pipeline #5, #6 |
| **BR4** | When all openings are filled, the team is told to close the offer | pipeline #9 |
| **BR5** | A stage that still holds applications cannot be removed | pipeline #10 |
| **BR6** | Every move writes the history and queues the message in the same transaction | pipeline #3 |
| **BR7** | Closing an offer answers every candidate still in progress (“no candidate left without an answer”) | pipeline #14 |
| **BR8** | A pipeline has exactly one “hired” and one “rejected” final stage | pipeline #12 |

The core of the implementation is the function that moves an application; the extract below shows how the rules are applied inside one transaction:

```
// server/src/services/pipeline.js (extract)
export async function moveApplication(client, user, { applicationId, toStageId, reason }) {
  const { rows: [app] } = await client.query(`SELECT a.*, s.kind AS from_kind, j.company_id, ...
       FROM applications a JOIN stages s ON s.id = a.stage_id JOIN jobs j ON j.id = a.job_id
      WHERE a.id = $1 AND j.company_id = $2 FOR UPDATE OF a`, [applicationId, user.company_id]);
  if (!app) throw notFound('Application');
  await assertCanManageJob(client, user, app.job_id);                          // BR2
  const { rows: [target] } = await client.query(
    'SELECT * FROM stages WHERE id = $1 AND job_id = $2', [toStageId, app.job_id]);
  if (!target) throw new HttpError(400, 'This stage does not belong to the job'); // BR1
  if (target.kind === 'rejected' && !reason?.trim())
    throw new HttpError(400, 'A reason is required to reject a candidate');  // BR3
  await client.query('UPDATE applications SET stage_id = $2, stage_entered_at = now() ...');
  await client.query('INSERT INTO stage_history (...) VALUES (...)');        // BR6
  if (target.email_key) await queueCandidateMessage(client, { ...,
      delayHours: target.kind === 'rejected' ? config.rejectionDelayHours : 0 });
}
```

## 4.4 Detailed design

### *4.4.1 Data model*

The database has 13 tables (Figure 10). Three choices are important. The **phone number** identifies a candidate within a company (unique index): a person who applies twice, or whose CV arrives by WhatsApp and then online, is not duplicated. The **current stage** is a foreign key of the application, and every change is kept in `stage_history`, from which delays and the dashboard are computed. The **CV file** is never in the database: the table `cv_documents` stores only the key of the encrypted file, its SHA-256 fingerprint and the words of its text for the search.

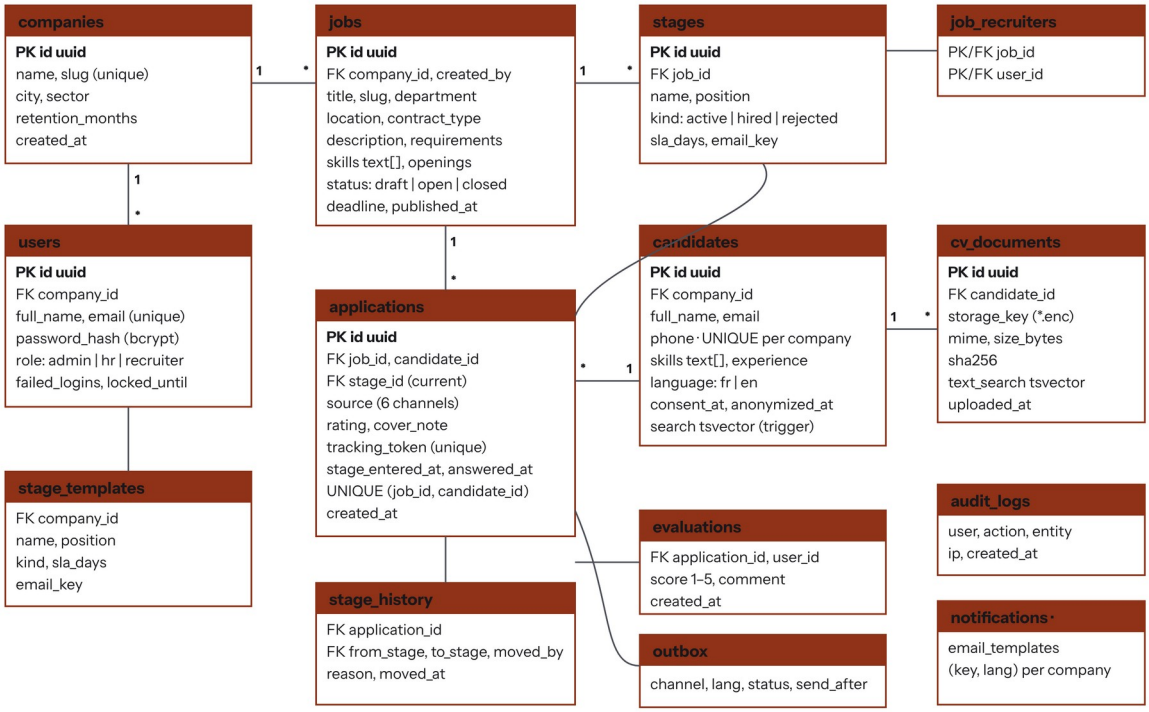



***Figure 10:*** *Physical data model (PostgreSQL) (source: author)*

### *4.4.2 REST API*

***Table 15:*** *Main REST routes (session required except the public routes)*

| Method | Route | Purpose | FR |
|---|---|---|---|
| POST | /api/auth/login, /logout | Open or close a session (httpOnly cookie) | FR-01 |
| GET | /api/public/:company/jobs[/:job] | Career page and offer | FR-06 |
| POST | /api/public/:company/jobs/:job/apply | Online application with CV (multipart) | FR-15 |
| GET | /api/public/track/:token | Public status of an application | FR-28 |
| GET / POST / PATCH | /api/jobs, /api/jobs/:id | List, create and edit offers | FR-05 |
| PUT | /api/jobs/:id/stages | Replace the stages of an offer | FR-10 |
| POST | /api/jobs/:id/publish, /close | Publish; close and answer everyone | FR-06, 07 |
| GET | /api/jobs/:id/pipeline | Kanban data | FR-11 |
| POST | /api/applications | Capture a CV received elsewhere | FR-16 |
| POST | /api/applications/:id/move, /evaluations | Move; evaluate | FR-11, 24 |
| GET | /api/candidates?q=&… | Search and filters | FR-22, 23 |
| POST / GET | /api/cv/:id/link, /api/cv/:id/file | Signed link; decrypted file | FR-18 |
| DELETE | /api/candidates/:id | Right to erasure | FR-32 |
| GET | /api/dashboard, /api/audit | Indicators; audit log | FR-30, 31 |

### *4.4.3 Secure management of CVs*

Figure 11 details the path of a CV, from the upload by the candidate to its consultation by HR. The measures and the tests that prove them are summarised in Table 16.

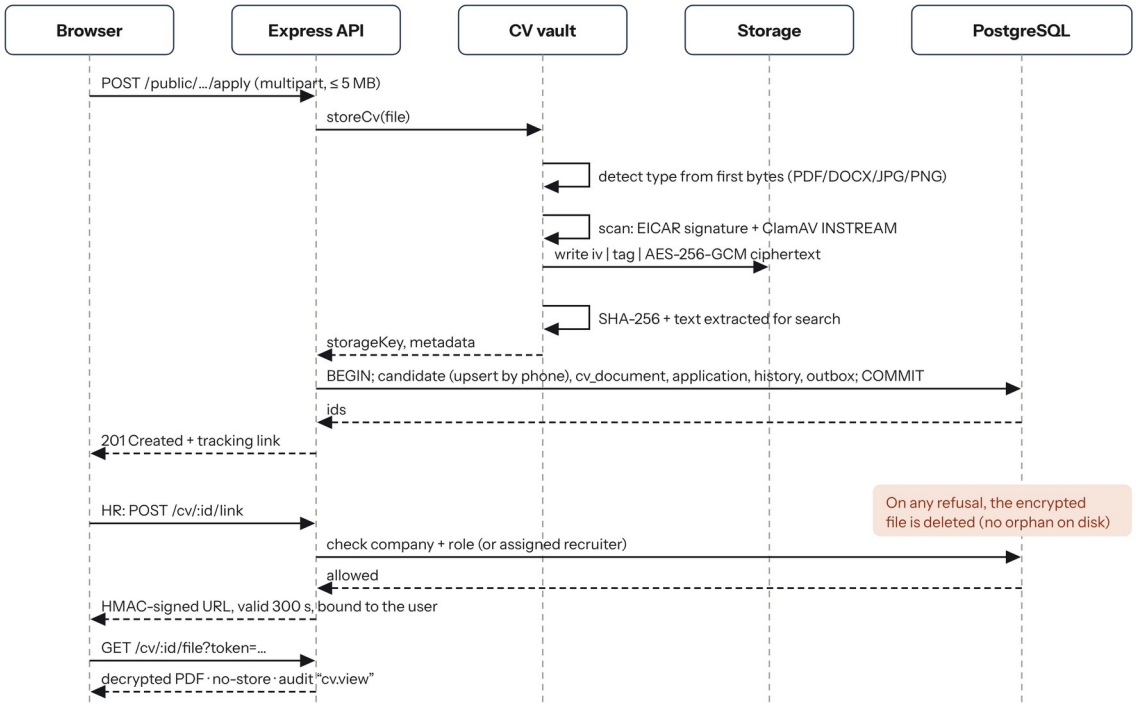



***Figure 11:*** *Sequence diagram of the secure upload and consultation of a CV (source: author)*

***Table 16:*** *Security measures applied to CVs and how they are verified*

| Threat | Measure | Verified by |
|---|---|---|
| Executable or script disguised as a CV | Real type read from the first bytes; allow-list PDF, DOCX, JPG, PNG; 5 MB limit | Tests: .exe renamed .pdf → 415; 5 MB + 10 bytes → 413 |
| Virus | EICAR signature check and ClamAV INSTREAM scan before storage | Test with the EICAR file → 422 |
| Theft of the disk or a backup | AES-256-GCM encryption, random name, key outside the database | Test: stored file unreadable; decryption gives the original |
| Unauthorised viewing | Company check, role or assignment; HMAC link valid 300 s, bound to the user | Tests: other company, unassigned recruiter, modified or expired link |
| Silent internal abuse | Audit log of every view; watermark with the viewer's name | Test: one line "cv.view" per opening |
| Brute force on accounts | bcrypt cost 12; 5 failures lock the account 15 min; rate limits | Test: sixth attempt → 423 |
| Excessive retention | Consent at application; erasure on request; anonymisation after 24 months | Tests: erasure deletes the files; retention job |
| Injection | Parameterised SQL everywhere; inputs validated by Zod | Test: " DROP TABLE " searched as text |

## 4.5 User interface design

A complete visual identity was created so that Bolo does not look like a generic template. The name comes from Cameroonian street French (“le bolo” = the job). The logo is a lowercase **b** whose bowl holds an ochre dot – the right person found among many (Figure 12). The main colour, **laterite**, recalls the red earth of Cameroonian roads; the login page uses a geometric pattern inspired by the ndop cloth of the Grassfields. Titles use the serif Fraunces and the interface uses Instrument Sans, both self-hosted (Figure 13, Table 17).




*Figure 12: The Bolo logo and its variants (source: author)*

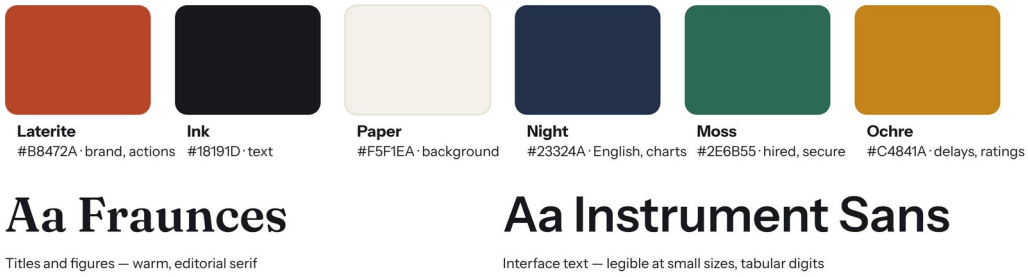



*Figure 13: Colours and fonts of the interface (source: author)*

*Table 17: Colour tokens of the interface*

| Token | Value | Role |
|---|---|---|
| **laterite** | #B8472A | Brand, primary buttons, active menu, French badge |
| **ink** | #18191D | Main text |
| **paper** | #F5F1EA | Background of the application |
| **night** | #23324A | English badge, charts |
| **moss** | #2E6B55 | Hired, security indicators |
| **ochre** | #C4841A | Delays exceeded, ratings |

Interface rules: the back-office is designed for a computer and the public pages **mobile first**; every candidate card shows the source channel, the language and the time spent in the stage, which turns ochre when the delay is exceeded; destructive or sensitive actions (rejecting, erasing) always ask for confirmation and explain the consequence.

## 4.6 Implementation

The application was built in three increments, each ending with automated tests and a commit. It contains about 3,500 lines of JavaScript, CSS and SQL in 53 source files (1,580 for the server, 1,420 for the interface and 480 for the tests). The screenshots below were taken on the running application with a **fictitious demonstration company**, “Wouri Distribution SARL”, and 171 fictitious applications created through the real services (each CV encrypted, each move written in the history).

**Increment 1 – accounts, offers and configurable stages (FR-01 to FR-14).** Login with a session cookie, roles, account lock; creation of offers with a copy of the company stages; stage editor; publication on the career page (Figures 14 to 16).




***Figure 14:*** *Login page of the HR workspace (source: author)*

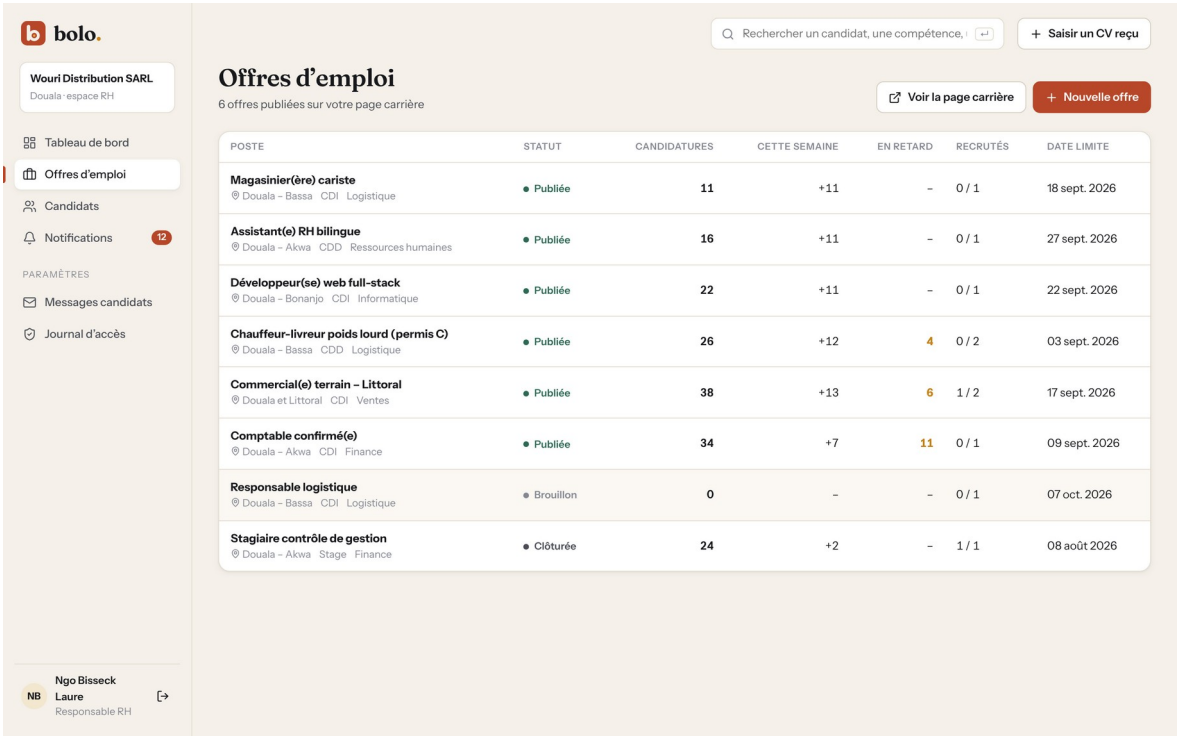



***Figure 15:*** *List of job offers with applications, delays exceeded and openings filled (source: author)*

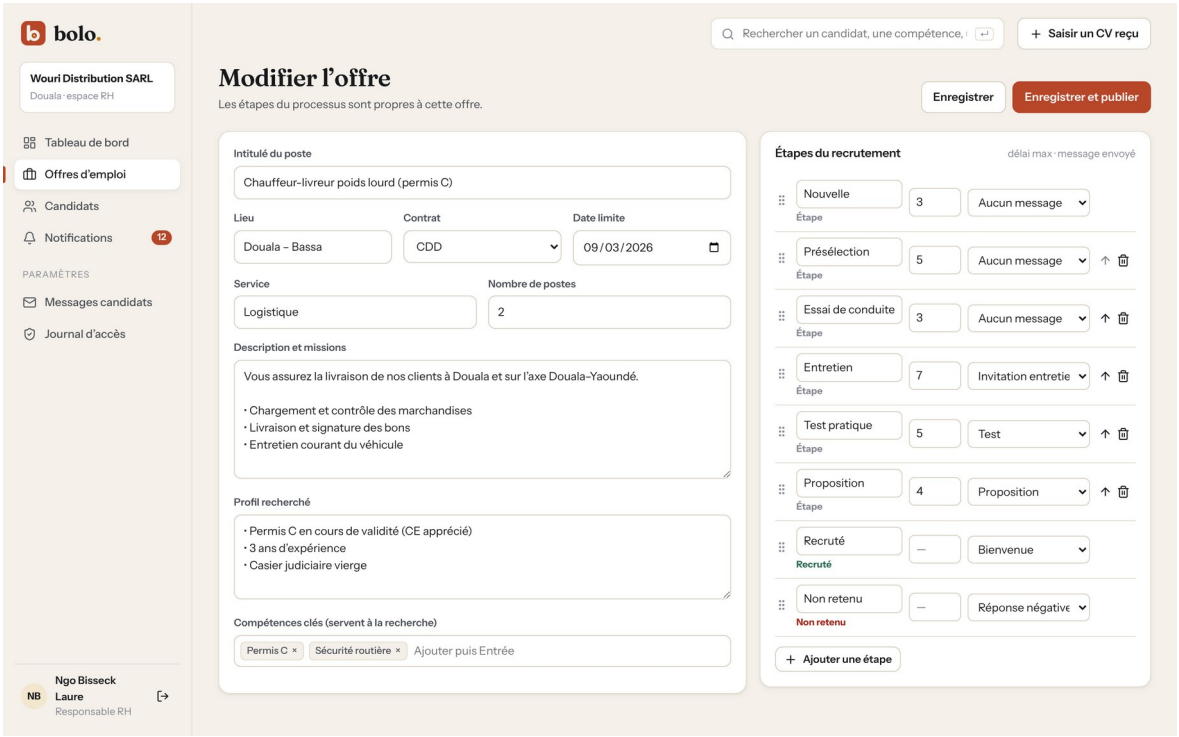
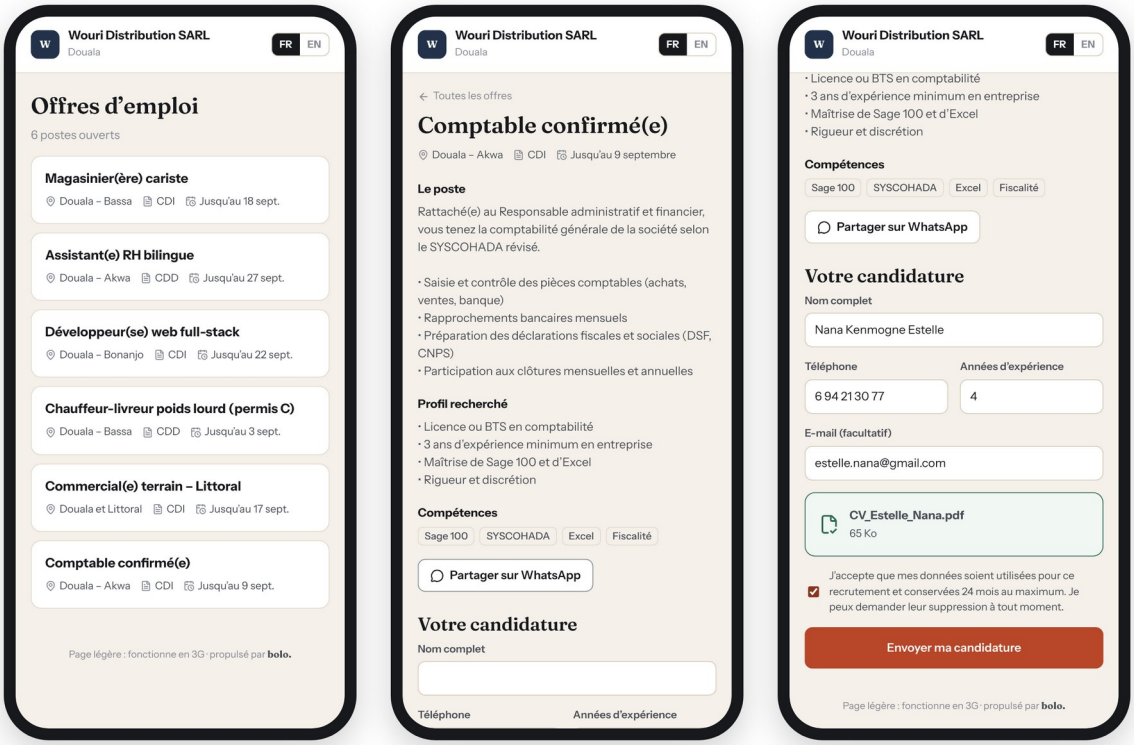



***Figure 16:*** *Editing an offer and its own stages (here with an added "Essai de conduite" stage) (source: author)*

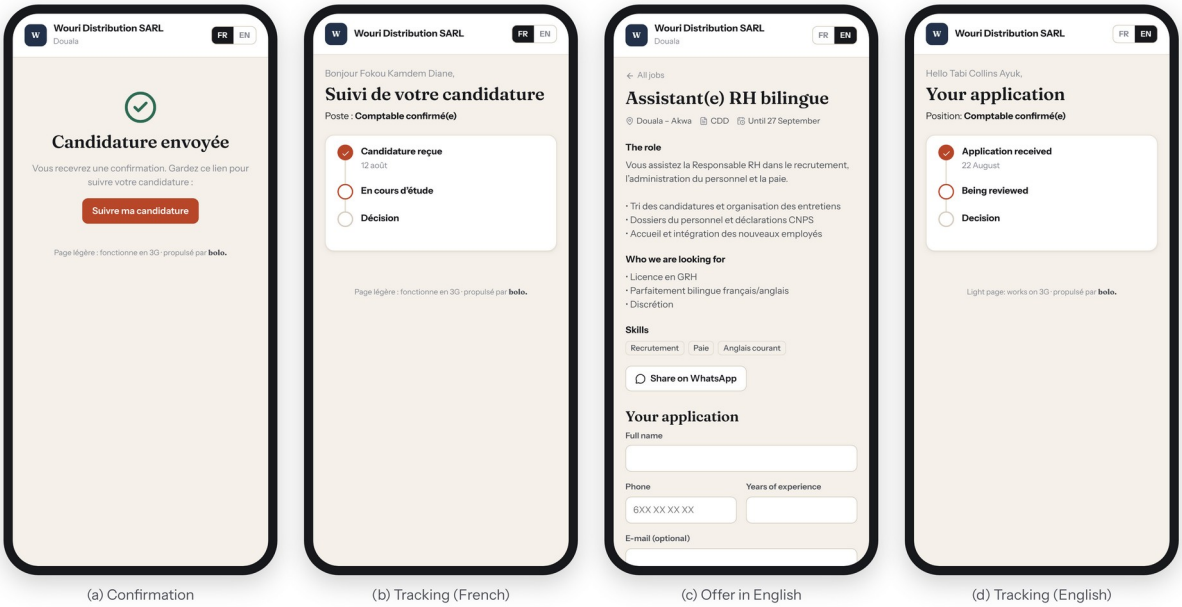
Problems solved: a JWT kept in the browser's storage could be read by a malicious script, so the session was moved to an **httpOnly, SameSite=strict cookie**; the default stages are copied – not shared – so that changing one offer never changes another.

**Increment 2 – applications and the CV vault (FR-15 to FR-21).** Public career pages in French and English, online application with CV or photo of a paper CV, manual capture of CVs received elsewhere, and the CV vault (Figures 17 to 19).




(a) Career page (b) Job offer (c) Application form

***Figure 17:*** *Public pages on a smartphone: (a) career page, (b) job offer, (c) application form (source: author)*




(a) Confirmation (b) Tracking (French) (c) Offer in English (d) Tracking (English)

***Figure 18:*** *After applying: (a) confirmation, (b) tracking in French, (c) offer and (d) tracking in English (source: author)*

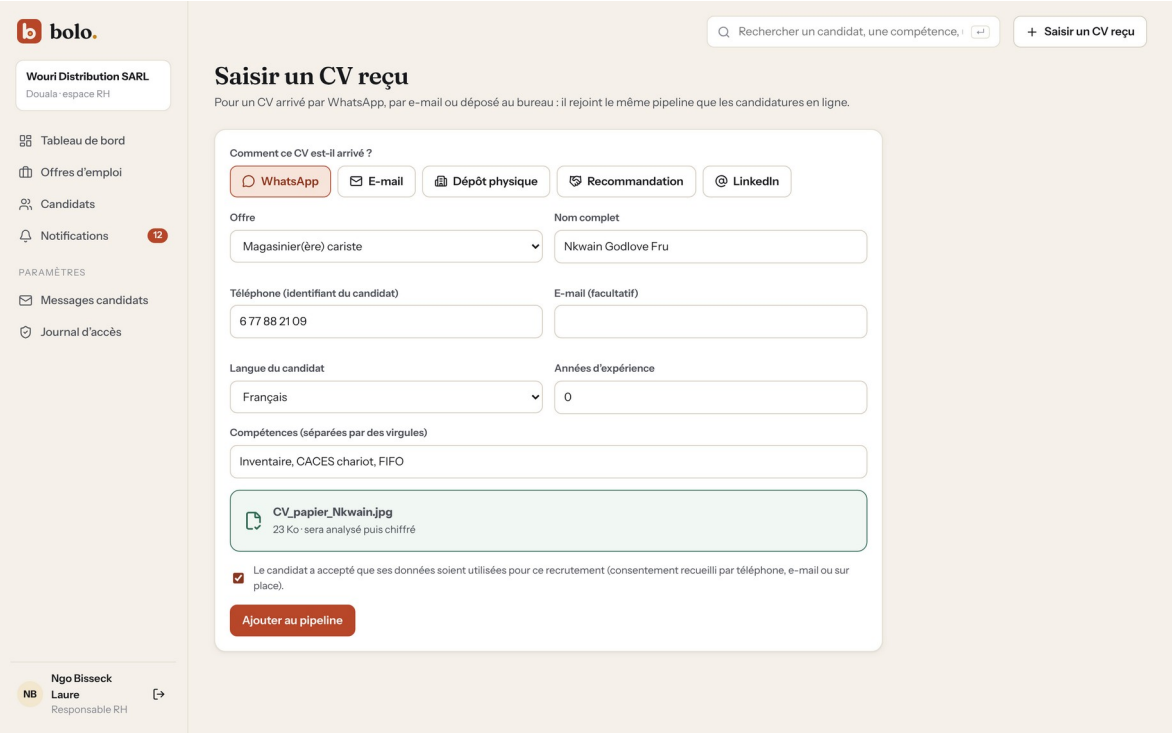



***Figure 19:*** *Capture of a CV received by WhatsApp (photo of a paper CV) (source: author)*

Problems solved: when an application was refused after the file had been stored, an encrypted file stayed on the disk; the upload is now undone in every error case (tested). A photo of a paper CV taken with a phone weighs 3 to 5 MB; it is now resized to 1,600 pixels and re-encoded in the browser before upload, which divides its weight by about ten for the candidate.

**Increment 3 – pipeline, search, messages and dashboard (FR-22 to FR-32).** Kanban board with drag and drop, rejection dialog, candidate profile with the secure CV viewer, search in profiles and CV text, bilingual message templates, notifications, audit log and dashboard (Figures 20 to 27).

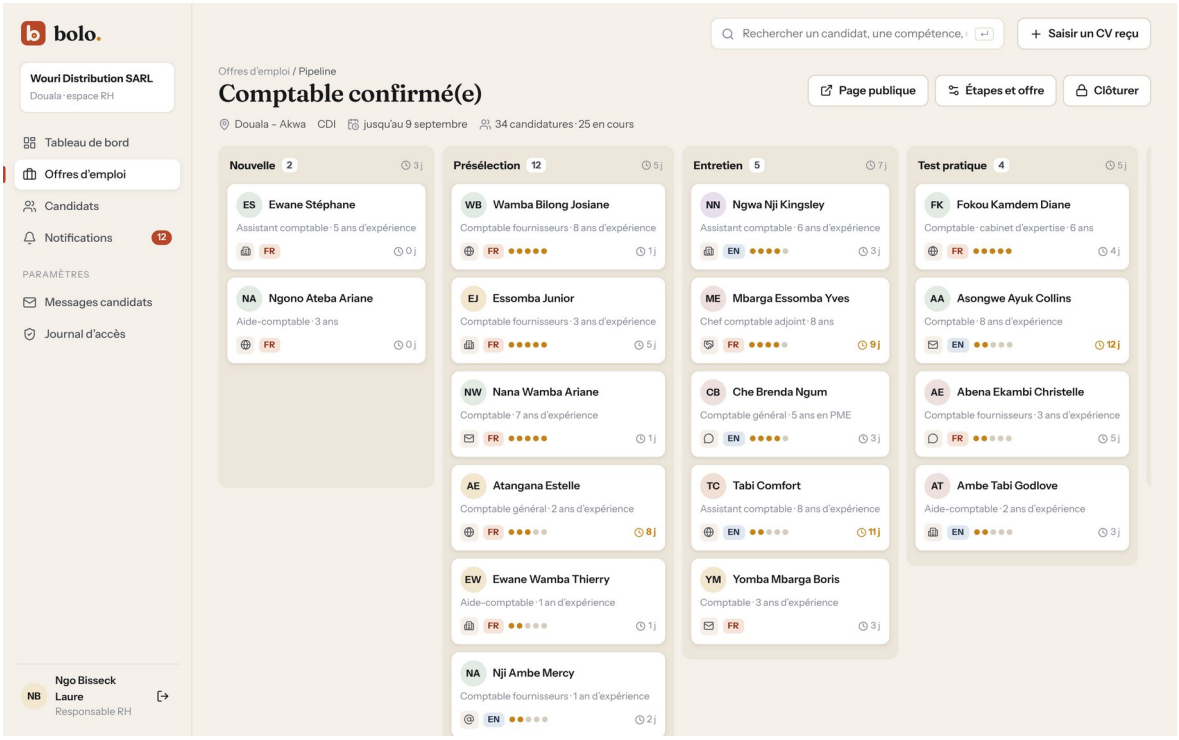



*Figure 20: Kanban board of an offer: source, language, rating and days in the stage on each card (source: author)*

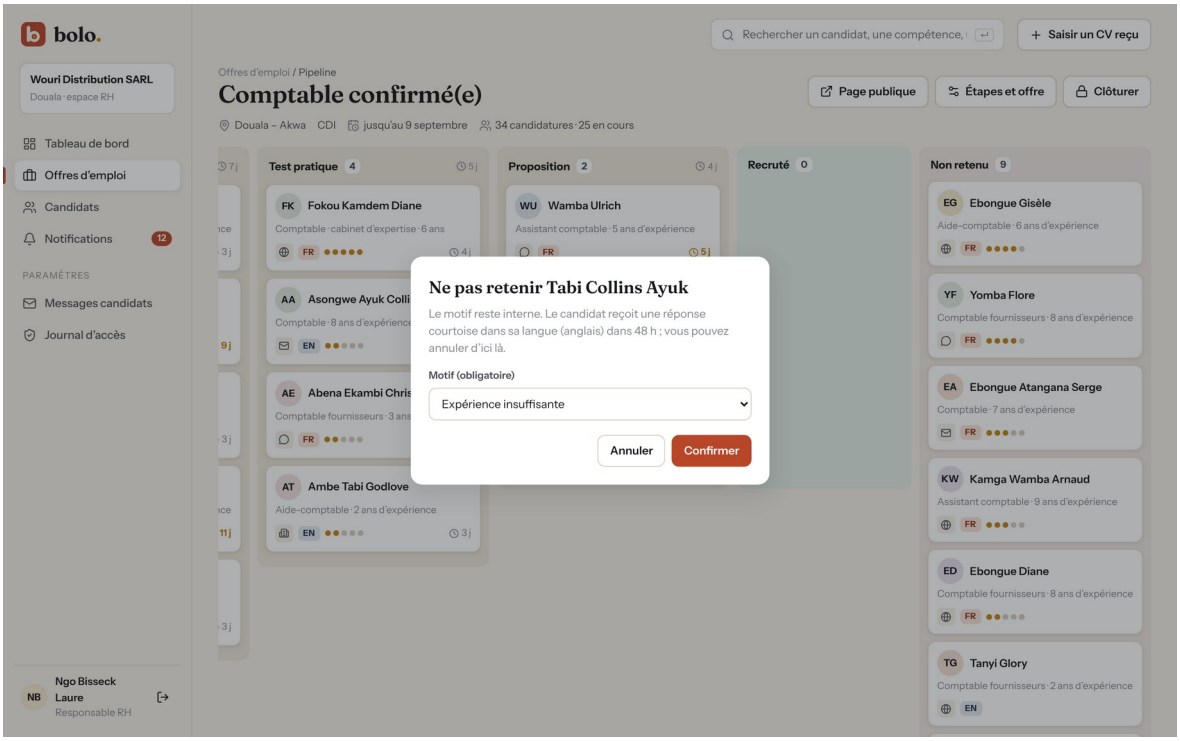



*Figure 21: Rejecting a candidate: the reason stays internal, the answer leaves 48 h later (source: author)*

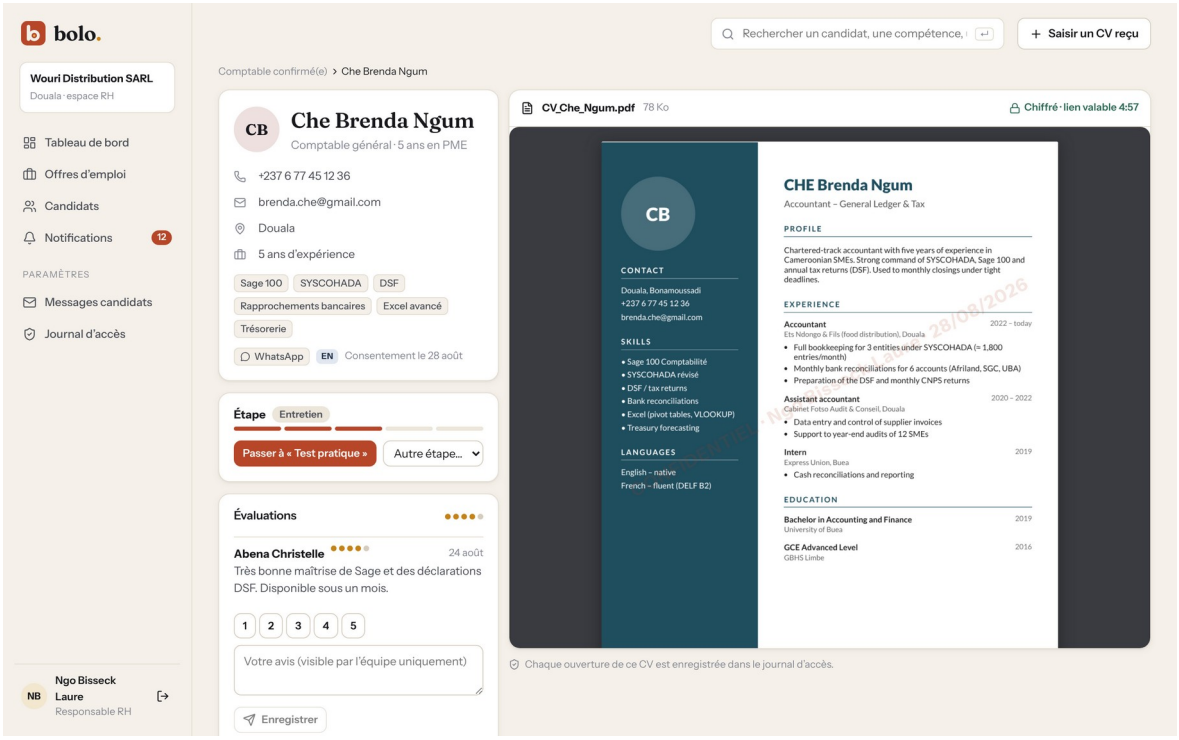



***Figure 22:*** *Candidate profile with the CV shown in the application, watermark and link countdown (source: author)*

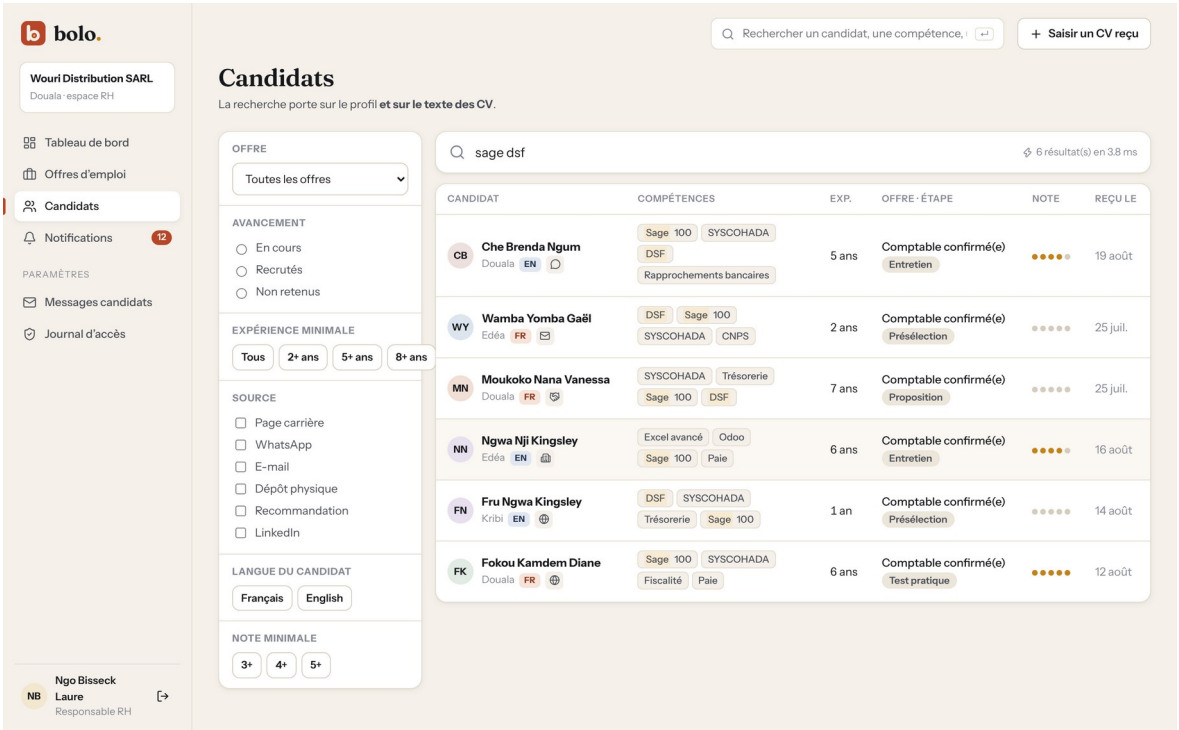



***Figure 23:*** *Search "sage dsf" in profiles and CV text, with filters (source: author)*

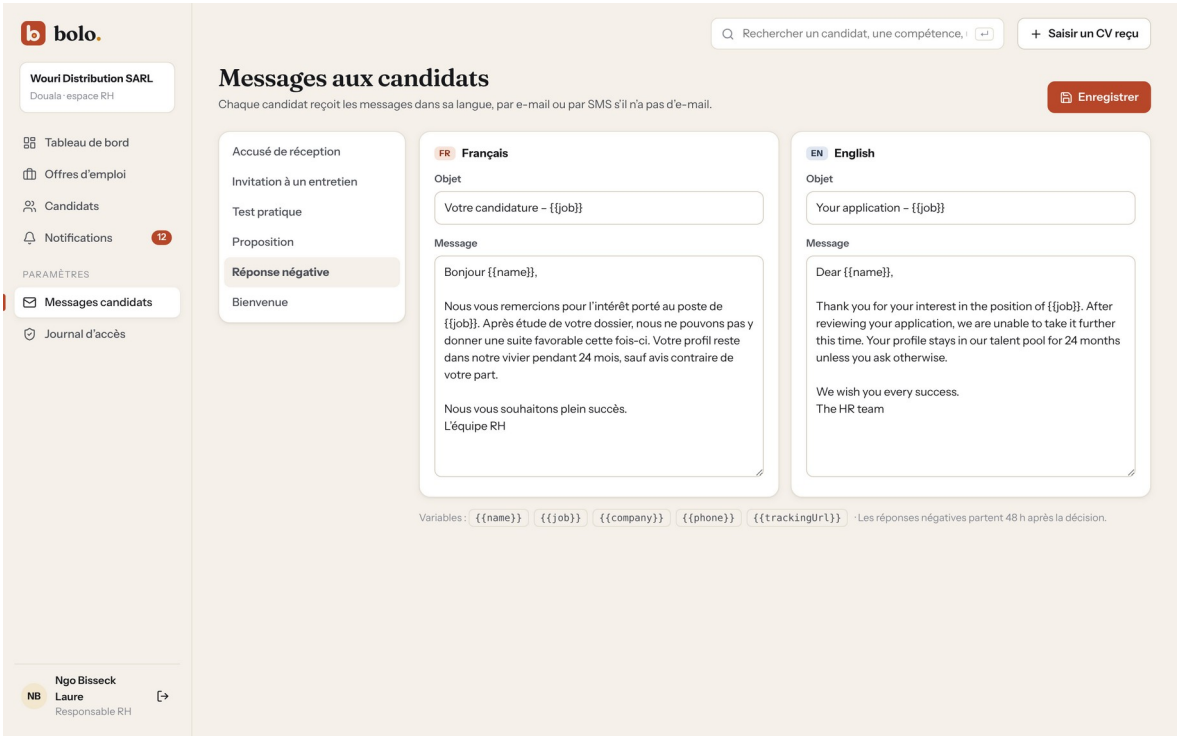



*Figure 24: Bilingual message templates sent to candidates (source: author)*

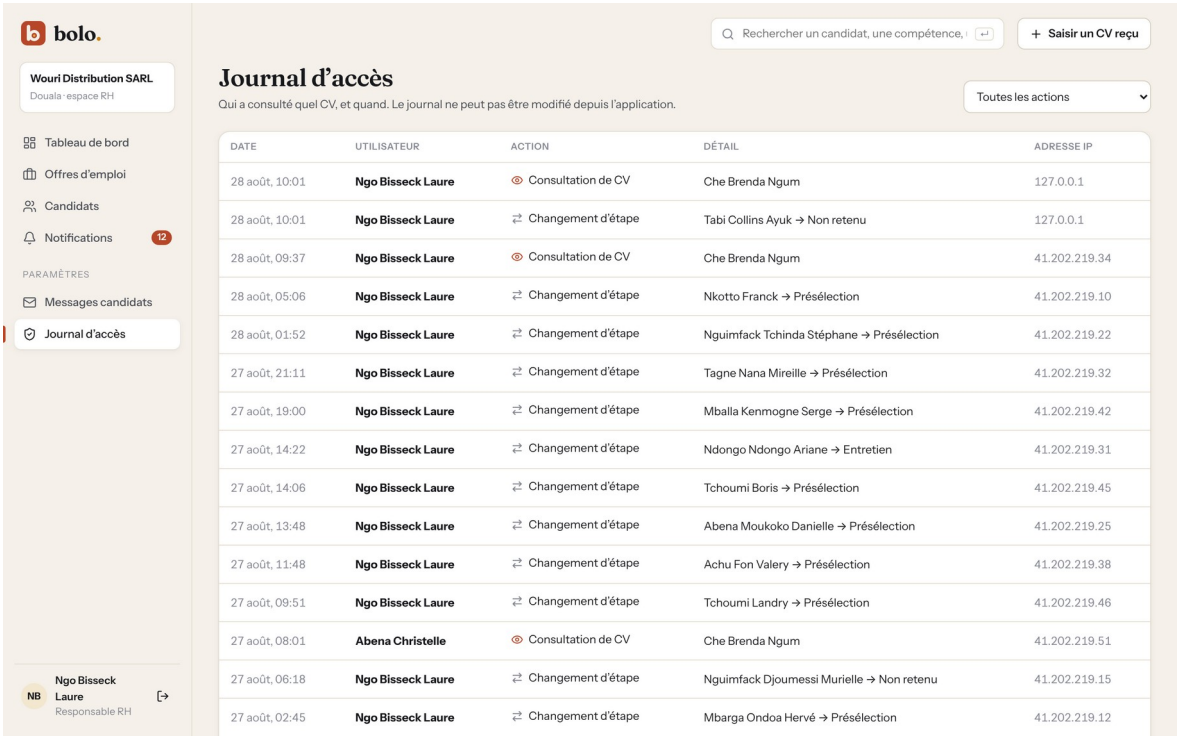



*Figure 25: Audit log: who viewed which CV, and when (source: author)*

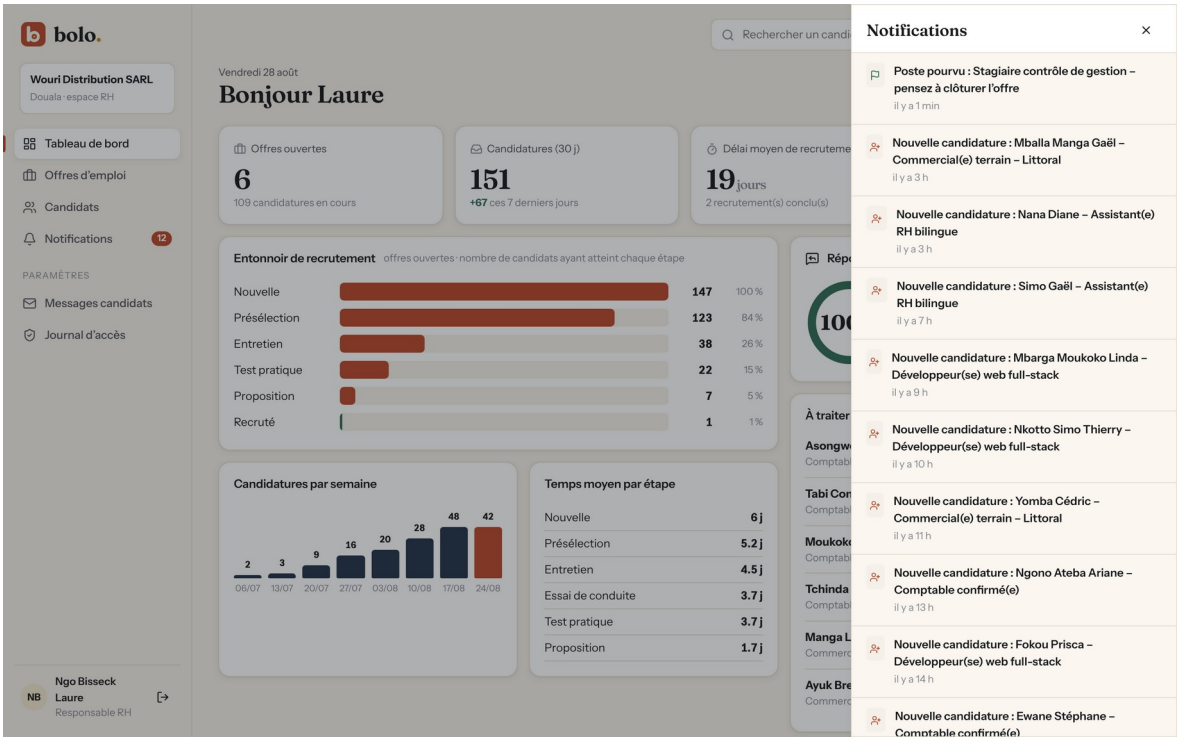



*Figure 26: In-app notifications of the HR team (source: author)*

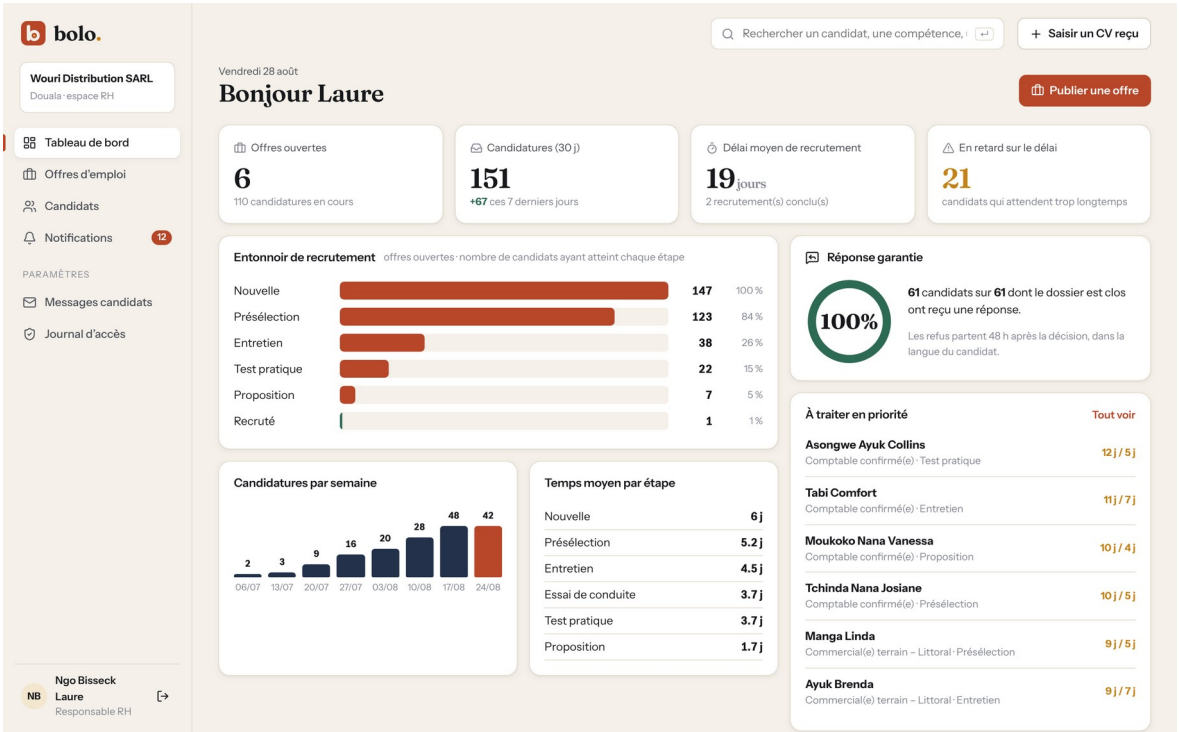



*Figure 27: HR dashboard: funnel, guaranteed answer rate, delays and sources (source: author)*

Problems solved: the search first missed “comptabilité” when the user typed “compta”; words are now searched **as prefixes** and accents are removed. A candidate rejected by mistake and moved back would still have received the rejection: the pending message is now **cancelled** when an application leaves a final stage. A test failed only because the database and the test runner did not use the same clock; time rules (account lock) are now evaluated by PostgreSQL alone.

## 4.7 Tests and results

Bolo is tested by **43 automated integration tests** (Node test runner and Supertest) that call the HTTP API against a real PostgreSQL 16 database, recreated before each test file with two companies, so that isolation between companies is also tested. All 43 tests pass (Figure 29).

*Table 18: Automated tests per area*

| Area | Examples of checks | Passed |
|---|---|---|
| **Authentication & roles** | Right and wrong password; lock after 5 failures; session required; invalid token | 5 / 5 |
| **Online application & CV checks** | PDF accepted, encrypted and acknowledged; no duplicate by phone; SMS when no e-mail; JPEG accepted; .exe, EICAR, > 5 MB refused; consent required; no orphan file; public status | 11 / 11 |
| **Pipeline business rules** | BR1 to BR8; recruiter scope; isolation between companies; evaluations; closing an offer | 14 / 14 |
| **CV access, search, erasure** | Signed link, modified and expired link, other company, unassigned recruiter; search in profile and in CV text; injection; erasure | 7 / 7 |
| **Messages, dashboard, retention** | Delayed sending; public status before and after the answer; custom and disabled templates; dashboard counts; roles; retention | 6 / 6 |
| **Total** | | **43 / 43** |

The performance requirements were measured with a script (`server/scripts/bench.js`) on a database of **10,000 candidates** and 10,000 applications, on the development server (Figure 28).

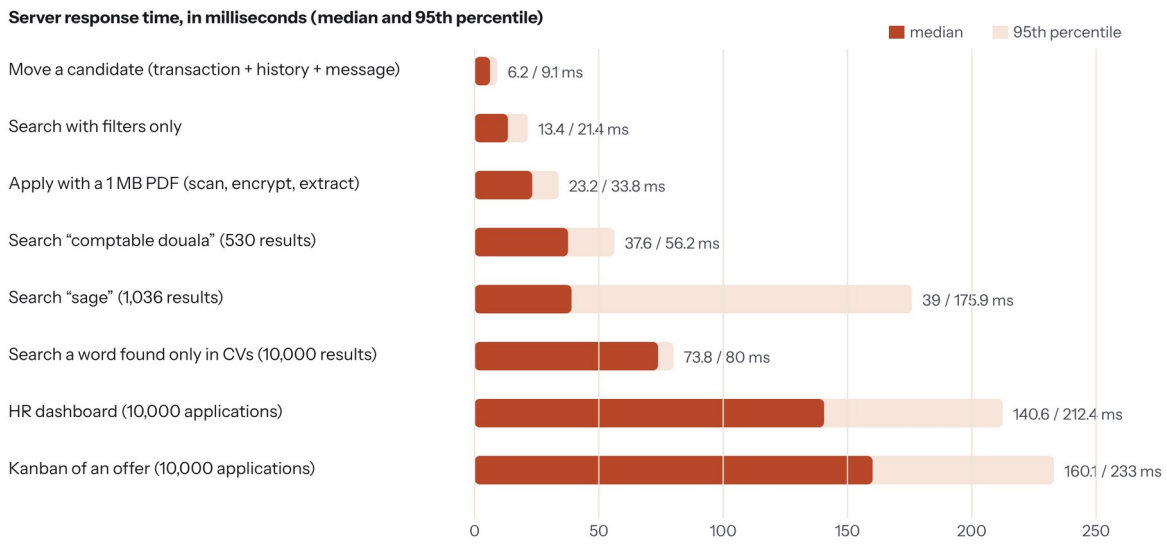



*Figure 28: Measured response times with 10,000 candidates (median / 95th percentile) (source: author)*

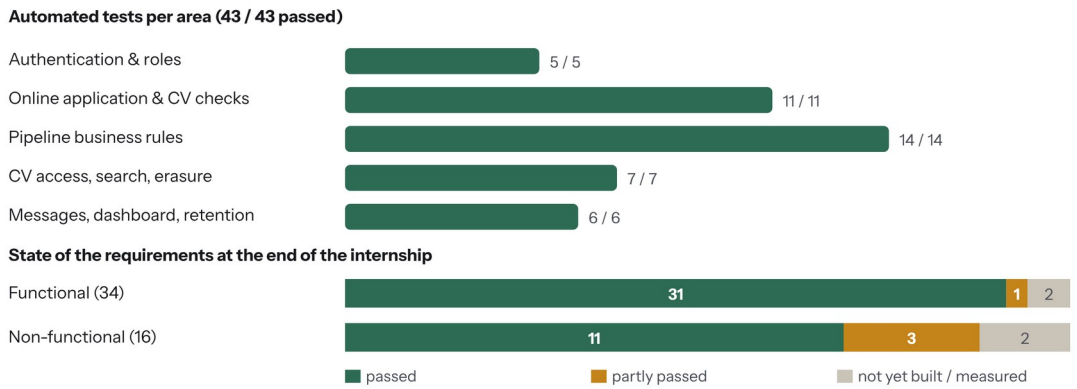



***Figure 29:*** *Automated tests and state of the requirements at the end of the internship (source: author)*

***Table 19:*** *Results of the non-functional requirements*

| ID | Target | Result | Status |
|---|---|---|---|
| **NFR-01** | Search 10,000 candidates < 1 s | Median 37–74 ms, worst 95th percentile 176 ms | Passed |
| **NFR-02** | Move < 300 ms | Median 6.2 ms, 95th percentile 9.1 ms | Passed |
| **NFR-03** | Career page < 300 kB; usable on 3G | 276 kB raw, about 136 kB with gzip; not yet measured on a real 3G network | Partly |
| **NFR-04** | 1 MB CV processed < 2 s | Median 23 ms on the server (network time not included) | Passed |
| **NFR-05 to 08** | bcrypt, lock, sessions and roles, isolation | Covered by tests and code review | Passed |
| **NFR-09** | Type, antivirus, encryption | Type and encryption tested; antivirus tested with EICAR only, ClamAV not yet run in production | Partly |
| **NFR-10, 11, 12** | Signed links, audit, privacy | Covered by tests | Passed |
| **NFR-13** | Offer published < 10 min; SUS ≥ 70 | Usability test with users planned (Appendix 1) | Not yet measured |
| **NFR-14** | Main browsers, from 360 px | Checked on Chromium (desktop and 390 px mobile); other browsers not yet tested | Partly |
| **NFR-15** | Layered code, tests | 3 layers; 43 tests; every business rule tested | Passed |
| **NFR-16** | One-command installation; backups | Docker Compose files written; production deployment not yet done | Not yet measured |

Among the 34 functional requirements, 31 are complete, one is partly done (FR-29: messages are produced, queued and “sent” by a console transport; the connection to a real SMTP server and SMS aggregator remains to be configured) and two optional ones are not built (FR-33 export to Excel and FR-34 interview scheduling).

## 4.8 Installation and configuration procedures

**Prerequisites:** a Linux server (2 vCPU, 4 GB of RAM), Docker and Docker Compose, a domain name and an e-mail account for sending. For development: Node.js 20 and PostgreSQL 16.

```
# 1. Get the code and create the configuration
git clone <repository> bolo && cd bolo
cp server/.env.example server/.env
openssl rand -hex 32        # -> JWT_SECRET
openssl rand -base64 32     # -> CV_ENCRYPTION_KEY (keep an offline copy!)
# 2. Start database, antivirus, API and web server
DB_PASSWORD=******** docker compose up -d --build
# 3. Development mode instead
cd server && npm install && npm run db:seed && npm run dev    # API on :5000
cd client && npm install && npm run dev                        # web on :5173
# 4. Tests
cd server && npm test                                          # 43 tests
```

The server reads its configuration from environment variables: `DATABASE_URL`, `JWT_SECRET`, `CV_ENCRYPTION_KEY`, `STORAGE_DIR`, `CV_MAX_BYTES` (5 MB), `CV_LINK_TTL_SECONDS` (300), `REJECTION_DELAY_HOURS` (48), `APPLY_LIMIT_PER_HOUR` (20) and `CLAMAV_HOST`. After installation, the administrator creates the accounts of the HR team, sets the default stages and adapts the message templates. The database is saved every night with `pg_dump`; the encrypted CV volume is saved with it, and the encryption key separately.

## 4.9 Cost estimate of the solution

*Table 20: Cost estimate of the solution (indicative)*

| Item | Quantity | Unit cost (FCFA) | Total (FCFA) |
|---|---|---|---|
| Software (React, Node.js, PostgreSQL, ClamAV, Docker) | Open source | 0 | 0 |
| Laptop computer | Already owned | 0 | 0 |
| Internet connection (data bundles) | 9 weeks | 5,000 | 45,000 |
| Server (VPS 2 vCPU, 4 GB) – first year | 12 months | 15,000 | 180,000 |
| Domain name (.cm) – first year | 1 | 25,000 | 25,000 |
| **Subtotal material and services** | | | **250,000** |
| Analysis, requirements and process modelling | 10 days | 25,000 | 250,000 |
| Design and visual identity | 5 days | 25,000 | 125,000 |
| Implementation (3 increments) | 20 days | 25,000 | 500,000 |
| Tests, measurements and documentation | 7 days | 25,000 | 175,000 |
| **Subtotal handwork** | 42 days | | **1,050,000** |
| **TOTAL (material + handwork)** | | | **1,300,000** |

The handwork is valued at an indicative daily rate for a junior developer. One installation can serve several SMEs (each company is isolated in the database), which divides the hosting cost; sending SMS would add a cost per message, paid to a local aggregator.

## 4.10 Limits of the current version

- Messages are queued and their content is tested, but they are not yet sent through a real SMTP server or SMS aggregator.

- The antivirus path was tested with the EICAR signature; ClamAV itself has not yet been run in a production environment.
- The usability test with real HR users and the SUS questionnaire have not yet been carried out; the screenshots use fictitious demonstration data.
- Not built in this version: export to Excel, interview scheduling, import of CVs directly from a WhatsApp Business number or a mailbox.
- Rate limits are kept in the memory of one server instance.

## Conclusion

Bolo combines a single entry point for every channel, a recruitment process modelled as a configurable state machine, and a CV vault that protects personal data from the upload to the consultation. Its business rules and security measures are validated by 43 automated tests, and its response times stay far below the targets on 10,000 candidates; real message sending, the production antivirus and the usability test are the next steps.

# GENERAL CONCLUSION

### *Summary of the problems solved*

This internship at Afriland First Bank, from July 1 to August 31, 2026, aimed to give small companies a way to stop losing track of applications. The study started from five problems of the existing practice: CVs scattered over many channels, no shared view of the process, no traceability, candidates left without answer and personal data stored without protection. **Bolo** answers them with a single entry point – a light bilingual career page and the capture of CVs received elsewhere – a pipeline whose stages are defined for each offer, an answer guaranteed to every candidate in their language, and a vault that protects every CV.

### *Methodology and solutions proposed*

The project followed the software development life cycle with an iterative and incremental model: interviews and collection of the existing documents, a requirements specification of 34 functional and 16 measurable non-functional requirements, a BPMN model of the process and a configurable state machine, a UML and database design, a visual identity, and three tested increments on React, Node.js/Express and PostgreSQL. The platform is validated by 43 automated integration tests, all passed, and by measurements on 10,000 candidates.

### *Difficulties encountered and solutions*

Making the process configurable without making the code fragile was the main design difficulty: storing the stages as data with explicit invariants (one “hired” and one “rejected” stage, no deletion of a non-empty stage) solved it. File security was new to me: I followed the OWASP recommendations step by step and wrote a test for each threat. Several subtle defects were found by the tests – an orphan file after a refused application, a rejection message that would still have left after a candidate was moved back, two clocks disagreeing – and were corrected. Finally, the MoSCoW priorities made it possible to deliver every “Must” requirement within nine weeks.

### *Perspectives*

In the near future: connect a real SMTP server and a Cameroonian SMS aggregator; deploy on a server and run ClamAV in production; measure the career page on a real 3G network; run the usability test with five HR users and the SUS questionnaire; build the export to Excel and the scheduling of interviews. In later versions: import of applications from a WhatsApp Business number and from a mailbox, extraction of skills from CVs to suggest the best matches, and a multi-company offer that the bank could propose to its SME customers.

Beyond the application, this internship taught me how a bank organises its information system, how to lead a complete project from a business need to a tested product, and the rigour of measuring and documenting what is really done. It confirmed my interest in software engineering applied to the realities of Cameroonian companies.

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## *Webography*


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3. OWASP Top 10: https://owasp.org/Top10/
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7. DLA Piper, Data protection laws of the world – Cameroon: https://www.dlapiperdataprotection.com/?c=CM

# APPENDIX

## *Appendix 1: Usability test scenario (5 HR users)*

The test will measure NFR-13 (an offer published in under 10 minutes) and the System Usability Scale (SUS ≥ 70). Each participant performs six tasks on a computer, then fills in the SUS questionnaire (ten statements rated 1 to 5; score = sum of the adjusted answers × 2.5).

*Table 21: Tasks of the usability test*

| No. | Task read to the participant | Success criterion |
|---|---|---|
| 1 | « Publiez une offre de magasinier avec une étape “Visite de l’entrepôt”. » | Offer visible on the career page |
| 2 | « Ajoutez le CV reçu ce matin sur WhatsApp. » | Application in the first stage, source WhatsApp |
| 3 | « Trouvez les candidats qui connaissent Sage. » | Search used, list displayed |
| 4 | « Invitez Che Brenda à un entretien. » | Moved to “Entretien”, message queued |
| 5 | « Ne retenez pas Tabi Collins. » | Reason chosen, answer scheduled |
| 6 | « Qui a consulté le CV de Che Brenda aujourd’hui ? » | Audit log opened and read |

## *Appendix 2: Extract of the database creation script*

```
CREATE TABLE stages (
  id        uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  job_id    uuid NOT NULL REFERENCES jobs(id) ON DELETE CASCADE,
  name      varchar(60) NOT NULL,
  position  smallint NOT NULL,
  kind      varchar(10) NOT NULL DEFAULT 'active' CHECK (kind IN ('active','hired','rejected')),
  sla_days  smallint,
  email_key varchar(20)
);
CREATE TABLE applications (
  id               uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  job_id           uuid NOT NULL REFERENCES jobs(id) ON DELETE CASCADE,
  candidate_id     uuid NOT NULL REFERENCES candidates(id) ON DELETE CASCADE,
  stage_id         uuid NOT NULL REFERENCES stages(id),
  source           varchar(12) NOT NULL CHECK (source IN ('career_page','whatsapp','email',
                                                          'walk_in','referral','linkedin')),
  tracking_token   varchar(48) NOT NULL UNIQUE,
  answered_at      timestamptz,
  stage_entered_at timestamptz NOT NULL DEFAULT now(),
  UNIQUE (job_id, candidate_id)
);
-- one candidate per phone number and company: the phone is the identifier everybody has
CREATE UNIQUE INDEX candidates_phone_uq ON candidates (company_id, phone);
```

## *Appendix 3: Additional screenshots*

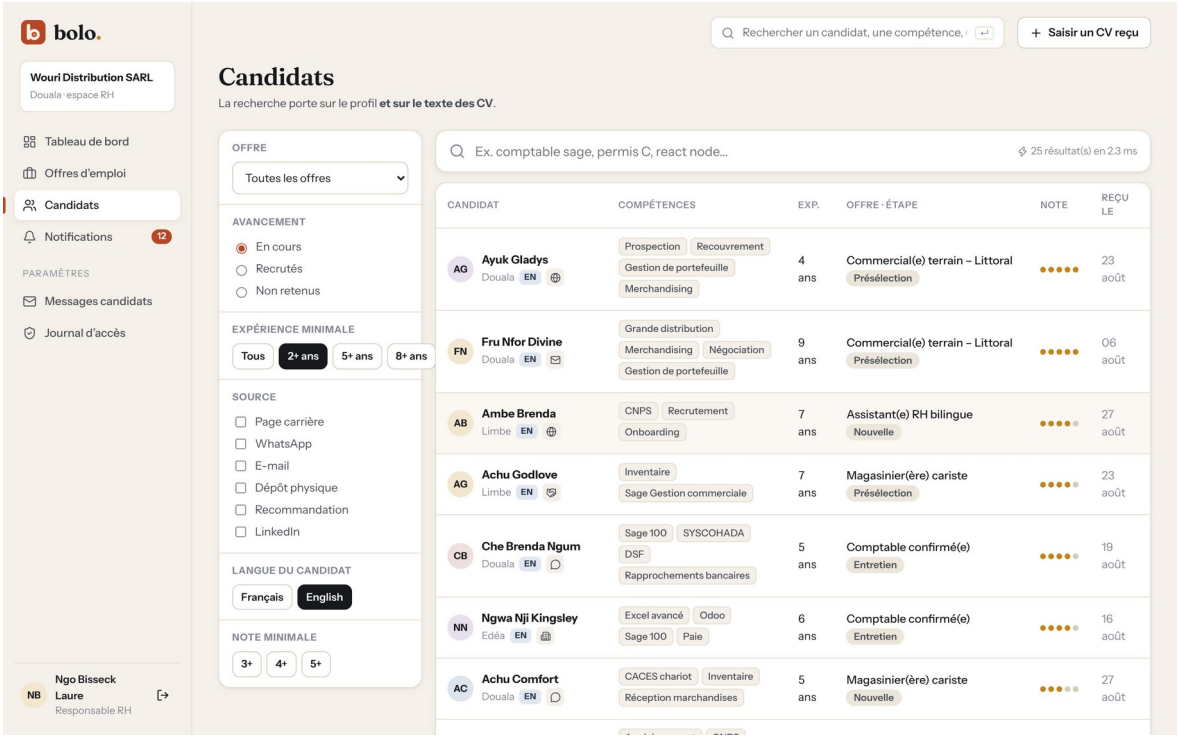



***Figure 30:*** *Filters: English-speaking candidates in progress with at least 2 years of experience (source: author)*

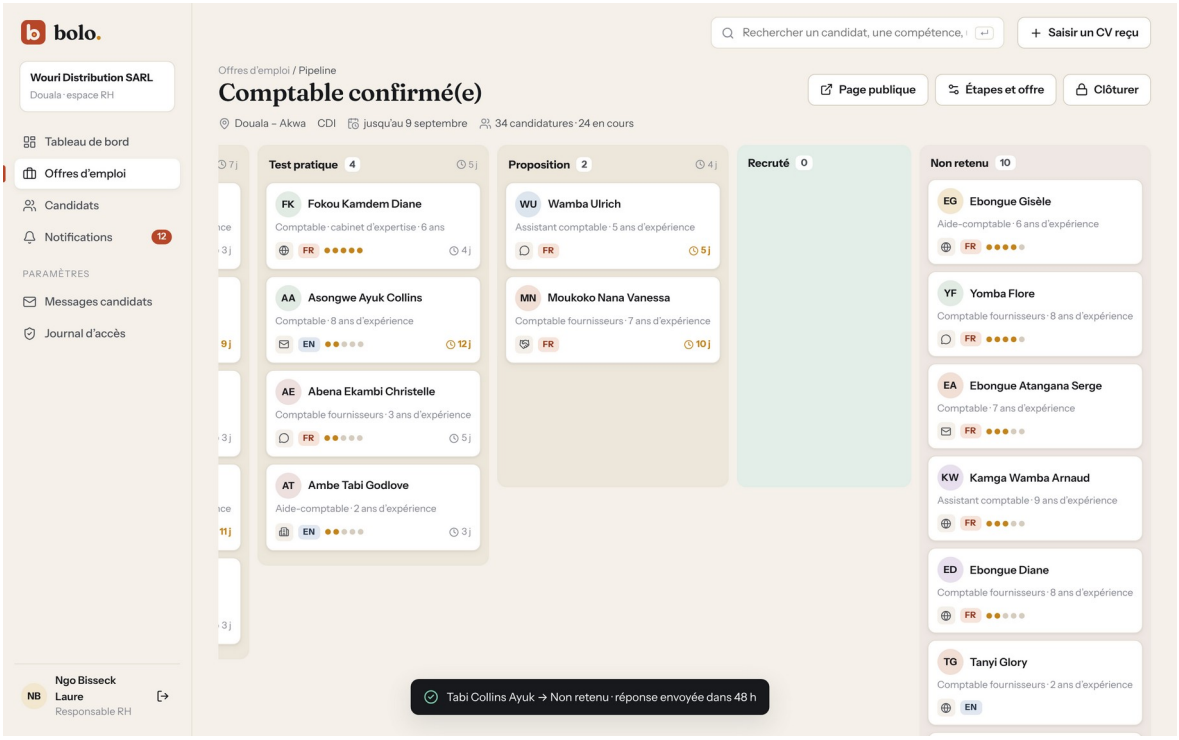



***Figure 31:*** *Kanban after a rejection, with the confirmation message (source: author)*